

Review

Digitalization for sustainable buildings: Technologies, applications, potential, and challenges

Muhammad Asif^{a,c,*}, Ghinwa Naeem^a, Muhammad Khalid^{b,c}

^a Architectural Engineering and Construction Management, King Fahd University of Petroleum & Minerals, Saudi Arabia

^b Electrical Engineering Department, King Fahd University of Petroleum and Minerals, Saudi Arabia

^c IRC for Sustainable Energy Systems, KFUPM, Dhahran, Saudi Arabia

ARTICLE INFO

Handling Editor: Jing Meng

Keywords:

AI
BIM
Buildings
Climate change
Digitalization
Energy efficiency
Renewable energy
Smart meters
Sustainability

ABSTRACT

In the fight against climate change, and to improve its sustainability standards, the building sector is trying to reduce its energy consumption and environmental footprint. Leveraging digitalization, in the wake of Industry 4.0 and 5.0, holds immense potential in revolutionizing the buildings and construction sector. Digitalization is transforming buildings throughout their life cycle encompassing design, construction, operation, and retrofitting/decommissioning stages. This study aims to analyze the role of digital technologies in sustainable buildings. In doing so, a systematic literature review is conducted to examine the current landscape of the application of 13 key digital technologies - sensors, Building information modeling (BIM), Building Management Systems (BMS), smart meters, 3D printing/additive manufacturing, robotics, big data analytics, Internet of Things (IoT), machine learning, artificial intelligence, digital twin, blockchain, and cyber security - in terms of their scope and prospects in the building sector. These technologies have been discussed considering their functions, applications in buildings, adoption and market trends, challenges, and the way forward. The findings of the study reveal that digital technologies are making buildings more integrated, flexible, energy-efficient, smart, and sustainable by optimizing resource utilization, enhancing operational efficiency, and minimizing environmental impact. For instance, sensors enable real-time data collection, BIM facilitates sustainability assessments, and AI enhances energy management. Further, digital technologies can lead to a 30%–50% improvement in energy intensity in the building sector by 2040. While digital technologies offer promising solutions in terms of improving sustainability standards in buildings, they also face several challenges such as high initial cost, privacy and security risks associated with data breaches and cyber-attacks, and lack of customer awareness and a skilled workforce. Conducive policies and regulations are critical for creating an enabling environment that supports the adoption of digital technologies.

1. Introduction

Global warming is the biggest threat facing mankind (Amaripadath et al., 2023). Global warming and subsequent climatic changes are leading to wide-ranging weather abnormalities and a pattern of more frequent and intense natural disasters including storms, flooding, droughts, and forest fires (Handbook of Energy and Environmental, 2022). To fight climate change and realize sustainable development, the world needs a paradigm shift across all sectors. Buildings are a crucial and integral part of any society's infrastructure, influencing its techno-economic and social fabric. Also, for their significant energy and environmental footprint, buildings play a critical role in the planet's

energy and environmental scenario. Accordingly, buildings are projected to be actively contributing to the fight against climate change and the zero-carbon energy transition. The International Energy Agency (IEA) sets out stringent targets for the buildings sector requiring a ban on the sale of fossil fuel boilers in the building sector; all new buildings to be zero carbon by 2050; 50% of existing building stock to be retrofitted to zero-carbon-ready levels by 2040 and more than 85% of buildings to be zero carbon by 2050 (Bouckaert et al., 2021).

Through technologies that collect and analyze data to improve the physical environment, the digital revolution can increase energy efficiency. Energy efficiency is being modernized through digitalization, which also raises its value. Digitalization offers energy efficiency benefits beyond those attainable when energy end-uses remain

* Corresponding author. Architectural Engineering Department, King Fahd University of Petroleum & Minerals, Saudi Arabia.

E-mail address: dr.m.asif@gmail.com (M. Asif).

<https://doi.org/10.1016/j.jclepro.2024.141814>

Received 11 September 2023; Received in revised form 5 March 2024; Accepted 16 March 2024

Available online 21 March 2024

0959-6526/© 2024 Elsevier Ltd. All rights reserved.

Nomenclature		FDD	Fault Detection and Diagnostics
AI	Artificial Intelligence	GHGs	Green House Gases
ANN	Artificial Neural Network	HVAC	Heating, Ventilation, Air conditioning
ASR	Automatic Speech Recognition	ISO	International Standards Organization
AEO6	6th ASEAN Energy Outlook	IEA	International Energy Agency
BMS	Building Management Systems	IPCC	International Panel on Climate Change
BIM	Building information modeling	IoT	Internet of Things
CAGR	Compound Annual Growth Rate	LR	Linear Regression
CCTV	Closed-Circuit Television	ML	Machine Learning
DNN	Deep Neural Network	NLP	Natural Language Processing
DT	Digital Twin	RDSS	Revamped Distribution Sector Scheme
EWS	Efficient World Scenario	UK	United Kingdom
		UNEP	United Nations Environment Program

disconnected by increasing the connectedness of the world’s structures, appliances, and equipment. Between 2017 and 2040, digitalization can cut the energy demand in the building sector by up to 10% while increasing the capacity for demand response by ten times (Energy Efficiency, 2019, 2019, 2019). Besides energy efficiency, there are many other potential aspects of optimizing digital technologies throughout the lifecycle of buildings like more accurate and integrated building design, enhanced project coordination and management, improved waste management, enhanced user comfort and security, improved productivity and management of building systems, and predictive maintenance.

Applications of digital technologies to improve buildings’ sustainability are being actively studied by the research community. Several digital technologies - including Building Information Modeling (BIM), IoT, big data, artificial intelligence, machine learning, blockchain, 3D printing, and robotics – have been investigated by researchers. Al-Ali, A.-R. et al. (Al-Ali et al., 2017), for example, proposed an energy management system that uses big data analytics and IoT to monitor and control the energy consumption of building systems and appliances. Moreover, the results highlighted a 30% reduction in energy consumption through this system. Zhang, F. et al. (Zhang et al., 2022a) studied integrated applications of BIM and AI like automated 3D modeling, fault detection & diagnosis, and material recognition that ultimately lead towards efficient and sustainable building design, construction, operation, and management. Razali, M.F. et al. (Razali et al., 2019) summarized the applications of BIM throughout the building lifecycle while highlighting the concerned benefits and challenges. Dörfler, K. et al. (Dö et al., 2022) explored the potential of optimizing mobile robots for 3D Printing. Olu-Ajayi, R. et al. (Olu-Ajayi et al., 2022) developed an energy predictive model through Deep Learning and Machine Learning techniques for residential buildings. Moreover, this study optimizes smart meter data for future energy consumption prediction. Shojaei, A. et al. (Shojaei et al., 2019) proposed a model using blockchain to handle sustainability-related data like energy consumption and carbon emissions in the built environment. Yevu, S.K. et al. (Yevu et al., 2021) conducted a comprehensive review of digital technologies’ application for supply chain and procurement processes in the built environment, identifying key research areas and future opportunities for advancing digitalization in construction supply chains. Kumar Singh, A. et al., (Kumar Singh et al., 2023) identified and analyzed barriers to the adoption of blockchain technology in sustainable construction projects using a novel hybrid methodology. Key findings of this study revealed critical barriers such as inadequacies in policy implementation, unawareness among customers, technology immaturity, market uncertainty, and accessibility issues.

It is, however, noted that the literature on the application of digital technologies in the building sector has some limitations. The existing studies, for example, mainly highlight the application of certain technologies over a specific phase of the building lifecycle. None of the

studies has elaborately and collectively discussed the applications of the range of digital technologies over the lifecycle of a building. Also, digitalization is rapidly evolving in terms of not only technological solutions but also applications in buildings. Given the broader digitalization trends, there is, thus, a greater and continuous need to more closely study the prospects of digital technologies in the building sector. It is important to understand the optimal utilization of existing as well as emerging digital technologies over the building lifecycle and how these can contribute to making buildings more sustainable. This study aims to bridge the gap in the literature by providing a comprehensive overview of digitalization in buildings in terms of technologies, their application, and prospects. The main objectives of this study are:

- Explore the contribution of digitalization to sustainable buildings
- Describe the technical principles and application of different digital technologies in buildings
- Examine the prospects of digital technologies in buildings in terms of opportunities and challenges

In terms of structure, the paper is organized into six main sections. After the Introduction Section, Section 2 discusses the energy and environmental footprint of the building sector. Section 3 briefly presents the concept and definite ions of digitalization. Section 4 discusses the application of digital technologies in buildings before reviewing the relevant literature on the subject. It also discusses different digital technologies in terms of their fundamentals and prospects. Section 5 discusses the prospects of digital technologies in terms of their adoption and market trends. It also highlights the challenges facing digitalization in buildings and recommends solutions. Finally, Section 6 presents the key conclusions of the study.

2. Research methodology

The study benefits from a systematic literature review (SLR) to examine the current landscape of the application of digital technologies in buildings. A total of 2797 studies were initially identified of which after rigorous criteria 168 studies were finally considered in this work as shown in the PRISMA flow diagram depicted in Fig. 1. Additionally, this study comprehensively reviews academic, scientific, and industrial literature, including journal and conference papers, books, market reports, and electronic articles from digital databases to identify important digital technologies for sustainable buildings and to examine their key aspects including functions, applications, adoption, market trends, and challenges. Given the rapid advancements in digital technologies, it also reviews grey literature including institutional and industry reports, newsletters, expert presentations, and national statistics. For the temporal scope, this study considered the literature published over the last decade, from 2013 to 2023. The graph in Fig. 2 depicts the yearly distribution of the utilized literature.

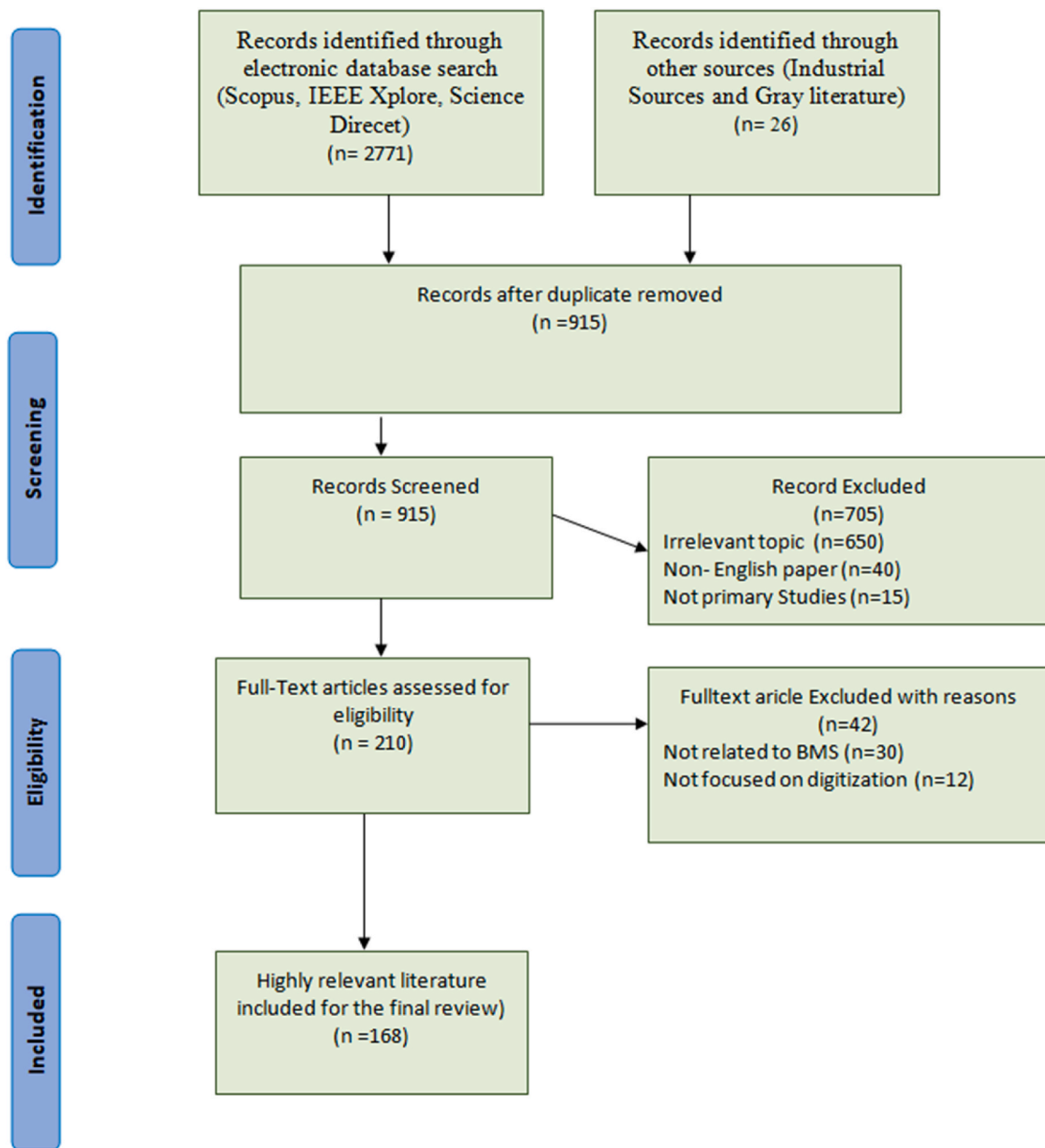


Fig. 1. PRISMA flow diagram for the Systematic Literature Review (SLR).

The study then discusses 13 key digital technologies considering factors like their relevance to buildings, prevalence in the industry, smart features, and potential impact on buildings' sustainability. Subsequently, the findings of the literature review are synthesized to identify the functionalities and applications of the selected digital technologies in the building sector also looking into factors like their impact on building integration and building sustainability and performance enhancement. It also discusses the market trends and challenges associated with digital technologies in buildings. Fig. 3 provides an overview of the literature selection process.

3. Digitalization: definitions and concept

Digitalization started around the 1950s with the advent of computers, sprouted in the 1980s with the invention of the internet and since then it has evolved rapidly through sensors, mobile technology, big data,

cloud computing, IoT, AI, and others, paving the way towards technological development and services (Njuguna Ndung'u, 2020). Digitalization is a disruptive concept, which refers to the utilization of digital technologies to transform or change business models, services, or products to create new value-producing opportunities and revenues (DigiFloat-Principle power). It refers to the expanding use of information and communication technology (ICT) in the economy, as stated by IEA" encompassing a range of digital technologies, concepts and trends such as artificial intelligence, the Fourth Industrial Revolution, and the IoT" (Frenzel et al., 2021). It involves the incorporation of digital technologies into daily life through the digitization of data. Digitization and Digitalization are closely related conceptual terms that are often used interchangeably but each term refers to a unique concept as Digitization is the conversion of any physical/non-digital object or analog to a digital representation. Digitalization on the other hand is referred to as the intelligent optimization of digital technologies and digitized data to

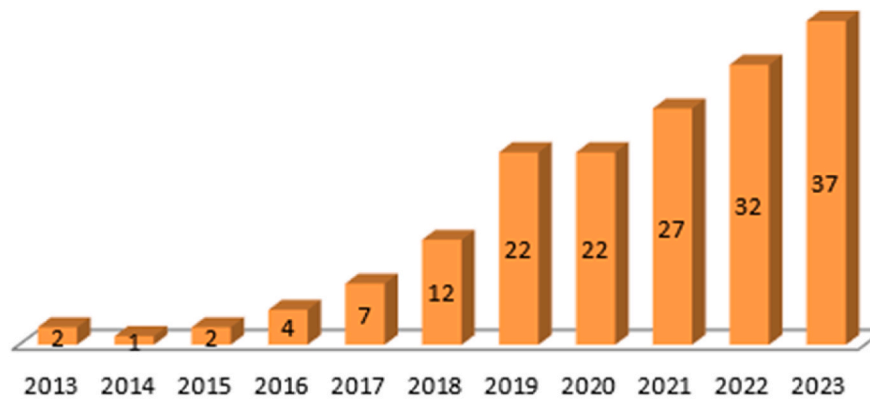


Fig. 2. Yearly distribution of the used literature.

enable or improve processes or services (Bumann et al., 2019). In the building industry, digitization includes the development of digital drawings and models, data collection through sensors, and optimization of software and common data environments to share and store data (Digitalisation in the construction sector, 2021). Broadly defined by the International Energy Agency (IEA); It is an evolving connection and emergence between the physical and digital worlds that hold immense potential to enhance the safety, productivity, efficiency, and sustainability of the system withholding the risks of security and privacy invasion (IEA, 2017). Although, the interrelationship between digitalization and sustainability in terms of its dimensions e.g., ecological, economic, and social is of critical importance as it varies with the actors' (policymakers, business leaders, civil society organizations, academics, and researchers) perceptions. For instance, Barbara Brenner & Barbara Hartl (Brenner et al., 2021) analyzed that some actors perceive that digitalization holds great potential to promote sustainability by lowering resource consumption, enhancing efficiency, and enabling new business models. However, some perceive that digitalization may impact negatively by increasing energy consumption and e-waste. Further, According to Sitsofe Kwame Yevu et al. (2021), digitalization may lead to more sustainable practices in the built environment by increasing environmental performance, reducing waste, increasing efficiency, and effectiveness, and improving value addition for stakeholders.

In the business world, digitalization is a commonly used and celebrated concept for which it is referred to as its transition to a digital business through the optimization of digital technologies and digitized data to transform the interaction of customers with companies and to create new revenues (Mangon, 2021; Merimi et al., 2021). In the context of buildings, Siemen describes digitalization as "Digitalization means building talk" (Ili, 2019) which highlights how digitalization makes it possible for buildings to exchange information and communicate, which can be utilized to design the best possible living and working environments. Hence, digitalization can be defined in various contexts but, it is the optimization of digital technologies and data to improve how we live, work, interact, connect, and function.

Digitalization is rapidly growing in the building sector (Digital transformation in the building, 2022). It is transforming the way buildings are designed, constructed, and operated. Building information modeling (BIM) is regarded as one of the important technologies driving the digital revolution in the building sector (Kaufmann et al., 2018). The rapid development and adoption of digital technologies have allowed building engineers, architects, and designers to utilize them throughout the lifecycle of buildings for efficient, sustainable, and smart building spaces.

4. Digital technologies for sustainable buildings

Buildings hold a significant share in the world's energy consumption

as well as environmental emissions (International Energy Agency, 2022; Minoli et al., 2017; Tricoire, 2021; Iqbal et al., 2019). Worldwide, the building sector accounts for over one-third of the total energy and materials consumption and releases around 40% of greenhouse gas (GHG) emissions (Sayadi et al., 2022; Qudrat-Ullah, 2020; Asif, 2022a). Buildings consume energy over their entire life i.e., construction, use, repairs, and decommissioning. (Alazazmeh et al., 2021). Of all the stages involved in this entire life cycle, the operational phase accounts for most of the energy used by buildings, and can range from 40% to 90% depending on factors like the nature of the building, user behavior, and weather conditions (Asif, 2022a). As per the 6th ASEAN Energy Outlook (AEO6), the energy consumption in Asia is presumed to be doubled from 2017 to 2040, with a two-third energy consumption increase in buildings by 2030 and 120% by 2040 (Christina Aprilia, 2022; Four ways smart building technology, 2021). Hence, Buildings' energy and environmental footprint are set to increase in the future as by 2050 the urban population is projected to increase from the 2018 level of 55%–70%. Subsequently, over this period around 2.5 trillion ft² of new floor area is expected to be added to the global building stock (Phillips et al., 2022). Given its massive share in the consumption of energy and natural resources, the building sector has a crucial contribution to make towards improving environmental sustainability and addressing the energy challenges (Ahmed et al., 2023; Asif et al., 2023; Alazazmeh et al., 2022). Moreover, the importance of buildings in improving global energy and environmental scenario has also been acknowledged in the United Nation's Sustainable Development Goals - besides, SDG 11 which targets sustainable cities and communities, several other SDGs are closely related to the built environment (Pandey et al., 2022). Recent reports by the International Panel on Climate Change (IPCC) conclude that meeting the climate change targets set under the Paris Agreement critically requires major improvements across four major global systems: energy, land use, cities, and industry (Asif, 2022b; Ahmed et al., 2021). Of these, energy, land use, and cities are directly and closely related to the building sector. The building sector has the greatest potential for providing long-term, significant, and affordable solutions to increase energy efficiency and reduce GHG emissions, as compared to other sectors like transportation and industry.

Therefore, there is a need to create a high-quality built environment that is productive, safe, resourceful, energy-efficient, and carbon-efficient and would give sustainable outcomes. Digitalization is one way of doing it (Digitalization - The Key Enabler of Smart, 2019) as it is revolutionizing the building industry by changing the ways, the building will be designed, constructed, operated, and managed, and buildings will be more networked, flexible, energy-efficient, smart, and sustainable (Magazine, 2022). For instance, the energy consumption of both new and existing buildings can be decreased by 30–80% using validated and commercially accessible technology, according to the United Nations Environment Program (UNEP), with a possible net profit

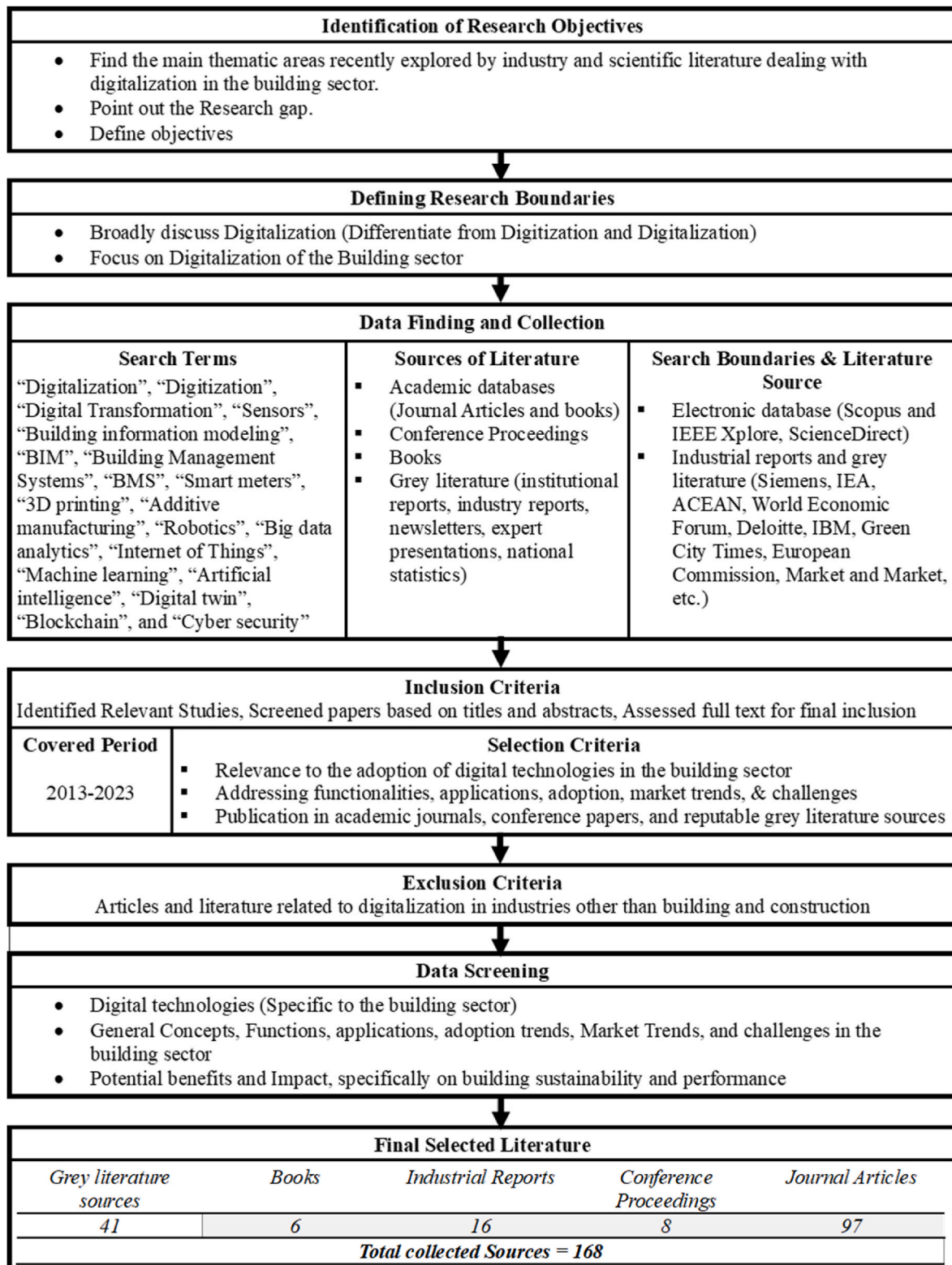


Fig. 3. Overview of the literature review process methodology.

throughout the building lifespan (Huovila et al., 2013). According to the International Energy Agency's (IEA's) Efficient World Scenario (EWS), the building industry might be 40% more efficient in 2040 than it was in 2017 thanks to new technology and policy directives. Buildings' potential Energy savings in many areas have been emphasized in Fig. 4 (Energy Efficiency, 2019, 2019, 2019). Hence, Digitalization and

sustainability are two of the important trends the building and construction industry is adopting. Digitalization promotes eco-innovation that ultimately contributes to sustainable development by minimizing environmental impacts and resource utilization. Moreover, digital technologies and data optimization encompass the potential to improve the building's sustainability by enabling energy efficiency, enhancing

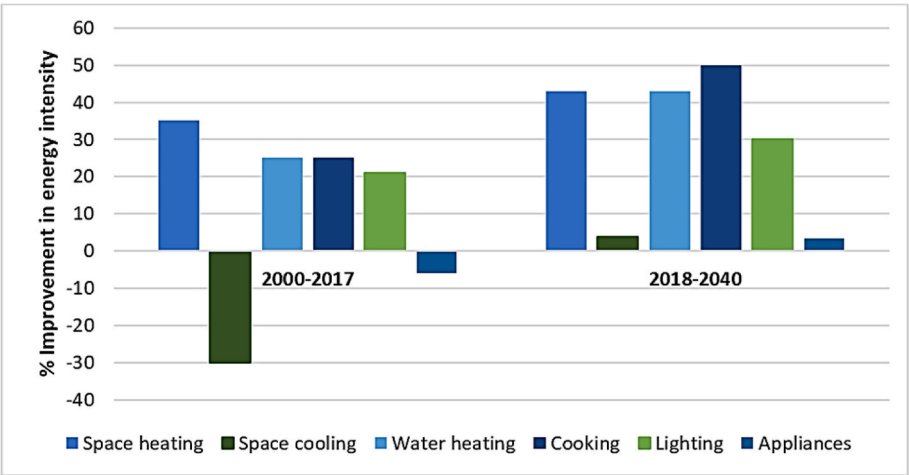


Fig. 4. Energy saving potential in buildings.

user comfort, limiting resource consumption, and minimizing CO₂ emissions throughout the building’s life cycle, while making the way towards the development of new services and markets to support the economical factor (Thelen et al., 2021).

The building industry is going through a major disruption due to

digital technologies – i.e., sensors, building information modeling (BIM), digital twin, internet of things (IoT), artificial intelligence (AI), machine learning (ML), 3D printing, robotics, smart meters, big data management, and blockchain - which can be used in buildings. The application of different digital technologies over the entire lifecycle of a building is

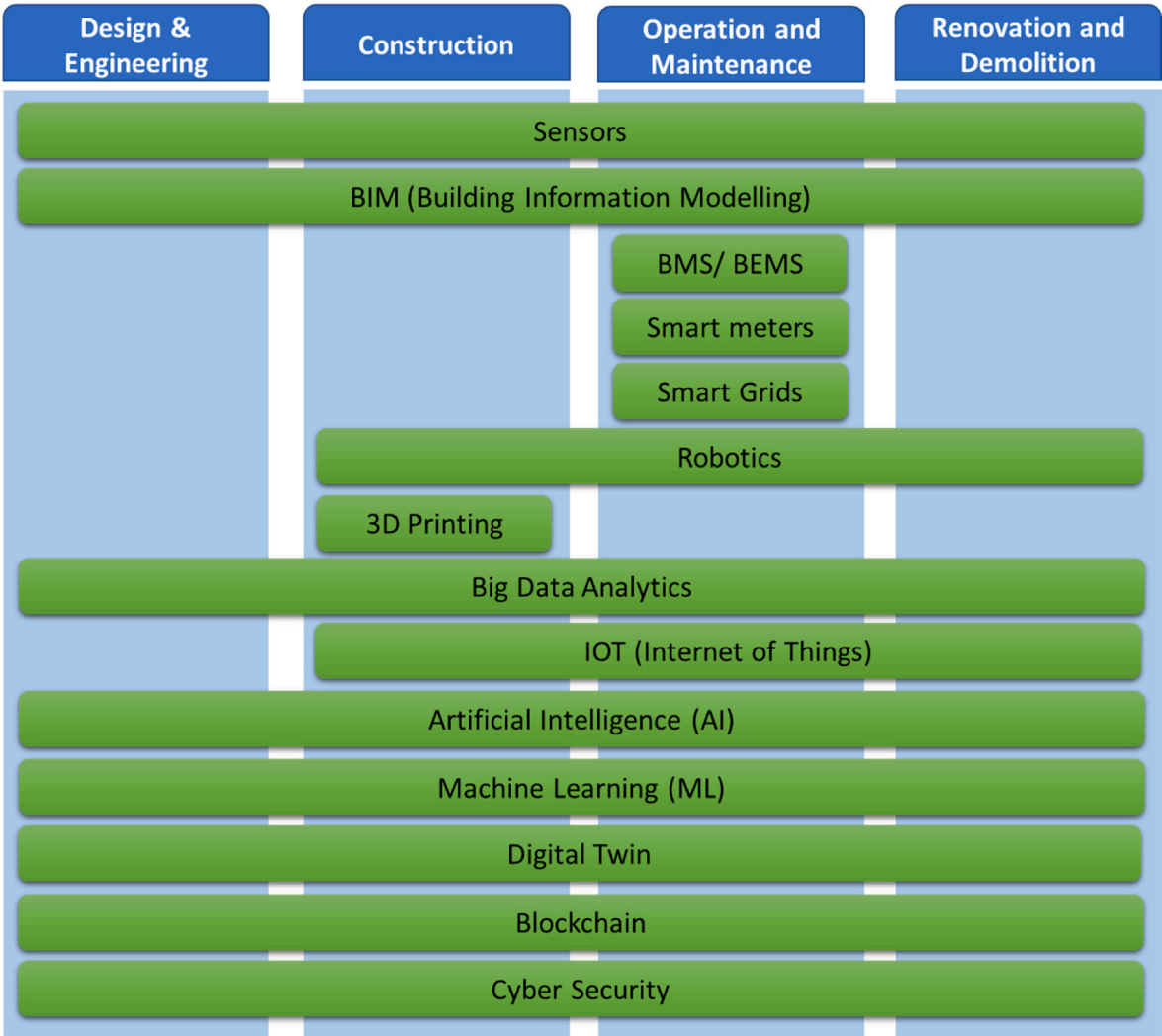


Fig. 5. Application domains of different digital technologies across a building’s lifecycle.

shown in Fig. 5.

These technologies hold the potential to enhance productivity, efficiency, safety, data sharing, waste reduction, and hence sustainability in buildings. 3D printing, for example, is considered to be more sustainable than conventional construction processes as it reduces cost, material waste, labor demand, and energy consumption in the construction phase which ultimately lowers product cost and carbon emissions (Malik, 2022). Fig. 6 provides a comparison of conventional and 3D printing construction techniques highlighting the significance of the latter in terms of energy and cost efficiency besides waste and emissions reductions (Singh et al., 2021). IoT and AI-enabled devices in buildings can facilitate managing and controlling indoor environmental parameters like indoor air quality thereby improving energy efficiency, safety, and comfort for occupants (Marzouk et al., 2022). BMSs are being employed in buildings for improved control and management of building systems that ultimately reduce energy consumption, and operation & maintenance costs, enhance user comfort, safety, indoor environmental quality, and productivity, and enable fault detection and emergency response, thereby paving the way towards sustainable built environment (Kumara et al., 2016). AI and ML technologies can improve a building's sustainability by providing real-time data collected from various sources like sensors, occupants' behavior, and weather data for predictive maintenance and to optimize energy consumption (Lyu et al., 2021). In the design and construction of buildings, BIM can aid in lowering material waste and increasing energy efficiency (Yevu et al., 2021). Blockchain technology can assist in increasing supply chain processes' traceability and transparency, which can improve resource management and lower carbon emissions (Yevu et al., 2021). To highlight the research trends, a review of pertinent studies on the application of digital technologies in buildings is furnished in Table 1. These papers have been selected in terms of their relevance to the research topic and their recency.

4.1. Overview of digital technologies in buildings

This section provides an overview of 13 prominent digital technologies in terms of their basic functions and applications in the building sector. The following digital technologies were selected after a comprehensive analysis of available literature and consideration of industrial prospects. Furthermore, the evaluation was done after careful consideration of their relevance and potential impact on the building industry while considering several factors e.g., market progression, adoption rates, and the substantial benefits they can offer to the building industry. For instance, sensors, IoT, 3D printing, robotics, drones, BIM, AI, ML, and digital twins are a few examples of digital technologies that can be used throughout or at any stage of the building's lifecycle, as stated by the European Commission (Digitalisation in the construction sector, 2021). Smart meters and BMS are essential for buildings' energy management thereby enhancing energy efficiency and sustainability

(Market and Market, 2023). Construction processes can be transformed by robotics and 3D printing (Alabbasi et al., 2023). The adoption of Big Data analytics, AI, ML, and IoT is growing to enhance building performance, occupants' comfort, and sustainability (Al-Ali et al., 2017; Zhao et al., 2020; Mehmood et al., 2019).

4.1.1. Sensors

Generally defined, a Sensor is a technical device that monitors the surrounding environment and generates signals in the form of analog at a constant frequency for data collection. In addition to gathering data, sensors can also send that data to computers and cloud networks for analysis, for instance through IoT systems to Digital Twins and BIM (Digitalisation in the construction sector, 2021) (Martin et al., 2021). A Physical sensor allows for the collection of data and the performance monitoring of specific types of information (such as power consumption, indoor air temperature, pressure, light, and CO2 concentration) for a building's lifecycle. A Virtual Sensor is a mathematical model or software that can predict and estimate desired physical variables, quantitative and qualitative performance indices, and operational building information that is difficult to measure with physical sensors due to some challenges in terms of measurability, precision, long-term measurements, and economic expenses in building operation (Yoon, 2022) (Choi et al., 2023). The sensor is the prime key to the digitalization of buildings (Yoon, 2022).

Several sensor types can be used throughout a building's lifecycle, for instance, temperature, pressure, humidity, motion/occupancy, contact, air quality, carbon, thermal, and electrical current monitoring sensors (Pradeep et al., 2021). During the construction phase of buildings, sensors can benefit from safety, optimization, and effective management as they can be integrated with drones to monitor the construction site environment. Sensors embedded in machines, allow the operator to know when maintenance is required thereby eliminating the repair cost. Moreover, it can be integrated with personal protective equipment to prevent safety and health risks for the workers. In the context of building operation and maintenance, it can ease the transformation from planned maintenance to predictive maintenance. It can control and monitor the building's internal and external environmental factors and operational equipment and systems like HVACs, lighting, alarm, and security systems (Digitalisation in the construction sector, 2021) (Pradeep et al., 2021).

4.1.2. BIM

Building Information Modeling (BIM), the term first appeared in 1992 has immensely evolved and is defined as the process of developing an intelligent virtual model that spans the entire lifecycle of a building from design and construction to operation and maintenance, by enabling efficient data exchange, collaboration, and management in digital format through a cloud-based shared platform that is "common data environment" (Chen et al., 2022) (Mangon, 2021). BIM is not just

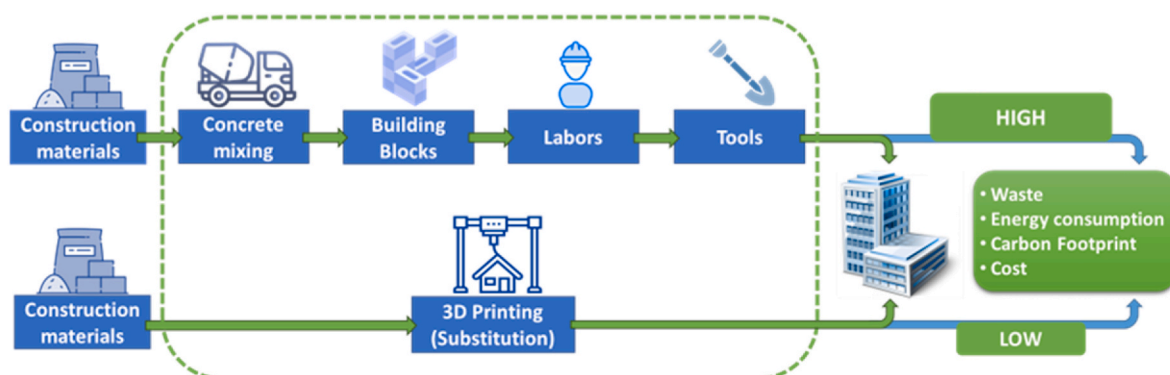


Fig. 6. Conventional vs. 3D printing construction processes.

Table 1

Applications of Digital technologies available in the literature.

Technology	Title	Description	Year	Ref.
DT	Digital Twin: Vision, Benefits, Boundaries, and Creation for Buildings	Provides an experimental analysis of DT application for life cycle management of buildings while analyzing the pros and cons of this implementation.	2019	Khajavi et al. (2019)
DT	A Digital-Twin Evaluation of Net Zero Energy Building for Existing Buildings	- Analyzed issues concerned with the creation of the building's DT, while opting for a building façade for experimental analysis. - Provides a future model for NZEB applications in existing buildings, which applies renewable technologies to the building by aiming to identify buildings' effectiveness and efficiency in energy consumption. - It infers that DT is proven to be feasible for all renewable technologies applied on the NZEBs	2019	Kaewunruen et al. (2018)
BIM	A Review: Application of Building Information Modeling (BIM) over Building Life Cycles	- Provides an analytical review of potential applications of BIM for buildings while outlining the challenges and prospects of future research. - A total of "108" papers, articles, and other academic sources were analyzed following the research focus.	2019	Razali et al. (2019)
BIM Robotics	BIM-Integrated Collaborative Robotics for Application in Building Construction and Maintenance	- Proposed a method to use BIM for the deployment of robotic systems during building construction and operation. - It infers that BIM provides Semantic and geometric data for robotic systems to function.	2021	Follini et al. (2020)
Robotics	A proposed ontology to support the hardware design of building inspection robot systems	Provides a conceptual framework for organizing complex building and defect information that helps to determine hardware design requirements for mobile building inspection robots (RoboDesign).	2023	Ma et al. (2023)
BIM AI	Integrated applications of building information modeling and artificial intelligence techniques in the AEC/FM industry	- Provides an analytical review of BIM and AI integrative approach (BIM-AI) for the AEC industry. - A total of 183 literature sources were analyzed for BIM-AI through systematic bibliometric analysis. - Potential applications for BIM-AI were analyzed over the building lifecycle following the automatic modeling and sustainable development	2022	Zhang et al. (2022b)
IoT	Development of an IoT assessment method: An interdisciplinary framework for energy-efficient buildings	- Developed a criteria framework to assess the IoT implementation level and the impact of its applications on buildings. - It signifies that IoT possesses the ability to integrate digital, and sustainable solutions	2022	Metwally et al. (2022)
IoT SS	An Energy-based Approach to Evaluate the Effectiveness of Integrating IoT-based Sensing Systems into Smart Buildings	Proposed Wireless sensing systems based on the IoT into smart buildings to provide insights into the overall sustainability and energy performance of buildings	2022	Kumar et al. (2022)
IoT AI	Intelligent Thermal Comfort Controlling System for Buildings Based on IoT and AI	- Proposed IoT and AI-based intelligent thermal comfort control system to improve IEQ and energy efficiency. - It reflects that the proposed system performed better than traditional control on comfort and energy savings.	2020	Zhao et al. (2020)
AI	Research on sustainability evaluation of green building engineering based on artificial intelligence and energy consumption	Proposed AI-based Energy Management Model (AI-EMM) to analyze energy usage data for green building and sustainable buildings which are responsive to human preferences, and make informed decisions to improve user comfort, safety, and energy efficiency.	2022	Xiang et al. (2022)
AI	Artificial Intelligence in green building	- Reviewed the adoption of AI in green buildings through bibliometric-systematic analysis method under areas; research trends, knowledge gaps, strengths, and limitations. - This mixed-methods systematic review is based on 383 research references that identify emerging AI technologies, techniques, and algorithms introduced in green building research.	2022	Debrah et al. (2022)
BT	Exploring the feasibility of blockchain technology as an infrastructure for improving built asset sustainability	A model is designed & evaluated, opting for BT as a tool to collect and distribute data to improve the sustainability of built assets at all stages	2019	Shojaei et al. (2019)
BT	Blockchain in the built environment: analyzing current applications and developing an emergent framework	- Investigated the application of BT in the built environment while developing a multi-dimensional framework for its application in the construction sector. - Identified its key application areas and summarized into seven categories: smart energy, smart cities and the sharing economy, smart government, smart home, intelligent transport, BIM and construction management, and business models and organizational structures. - It infers that the operational phase exhibited the most significant application of BT.	2018	Li et al. (2018)
BIM IoT BT	Application of Integrated Building Information Modeling, IoT, and Blockchain Technologies in System Design of a Smart Building	- Proposed and analyzed a framework for IoT and integrated BIM technologies while employing BT to control and secure it. - It highlighted that the integration of BIM and blockchain with provided IoT services in buildings can improve management and secure data during building operations.	2019	Lokshina et al. (2019)
BEMS	Multiple power-based building energy management systems for efficient management of building energy	- The proposed system design can be implemented in other types of buildings - Proposed a multi-power-based energy management system (MPBEMS) for effective energy management in buildings. - It highlights the significance of the proposed system for enhanced energy efficiency and energy consumption reduction.	2018	Yoon et al. (2018)
ML	Predicting energy consumption in multiple buildings using machine learning for improving energy efficiency and sustainability	- Proposed a Random Forests (RF) based prediction model for energy consumption in buildings. - The proposed RF model was analyzed to be the most effective as compared to other ML models that will help in comprehending building energy performance to improve building energy efficiency.	2020	Pham et al. (2020)

(continued on next page)

Table 1 (continued)

Technology	Title	Description	Year	Ref.
ML	Building energy consumption prediction for residential buildings using deep learning and other machine learning techniques	-Developed an annual energy consumption predictive model for the design stage of buildings using various ML techniques. -This study analyzed the prediction performance of nine ML techniques; Artificial Neural Network (ANN), Gradient Boosting (GB), Deep Neural Network (DNN), Random Forest (RF), Stacking, K Nearest Neighbor (KNN), Support Vector Machine (SVM), Decision tree (DT) and Linear Regression (LR) -It highlights DNN as the most efficient Prediction model thereby encouraging the usage of these models before construction to make an informed decision and manage and optimize the design.	2022	Olu-Ajayi et al. (2022)
VS	Virtual Sensors for Estimating District Heating Energy Consumption under Sensor Absences in a Residential Building	-Developed three virtual sensors through model-driven, data-driven, and grey-box approaches to analyze the energy consumption of district heating systems. -It reports that the data-driven virtual sensors exhibit more accurate predictions for energy consumption of District Heating as compared to others.	2020	Yoon et al. (2020)
VS DT	In-situ observation of virtual sensors in building systems toward virtual sensing-enabled digital twins	-Proposed a concept of in-situ observation virtual sensing enabled DT for building systems with an ability to measure and predict unmeasured variables in real-time. -It also inferred the advantages of reducing energy consumption thereby increasing the system's efficiency and occupants' comfort.	2023	Choi et al. (2023)
3D Printing BIM	3D Printing of Buildings: Construction of the Sustainable Houses of the Future by BIM	-It presented the potential of 3D printing for sustainable house construction while highlighting its significance for enhanced precision, efficiency, & sustainability.	2017	Sakin et al. (2017)
3D Printing	Robotic 3D printing of concrete building components for residential buildings in Saudi Arabia	-Presented a design to fabrication framework e.g., Robotic 3D printing, for the development of buildings' concrete components owing to the advantage of enhanced sustainability.	2023	Alabbasi et al. (2023)

DT: Digital Twin, **BIM:** Building Information Modeling, **AI:** Artificial Intelligence, **IoT:** Internet of Things, **SS:** Smart Sensors, **BT:** Blockchain Technology, **BEMS:** Building Energy Management System, **BMS:** Building Management System, **ML:** Machine Learning, **VS:** Virtual Sensors, **NZEB:** Net Zero Energy Buildings.

limited to 3D modeling, but it also encompasses various dimensions that allow the integration of useful data and information into BIM models like 4D (duration, scheduling, and sequencing), 5D (budgeting and cost), 6D (sustainability assessment and energy analysis) and 7D (facility management). However, the development of 6D in BIM is directly related to the increased awareness of sustainability in the building sector. Comprehensive assessment of a building's energy efficiency and environmental impact throughout a building's lifecycle is made possible by 6D BIM, which helps stakeholders make well-informed decisions for long-term sustainability while facilitating the integration of sustainable components such as renewable energy, daylighting, ventilation, water efficiency, and heat management. (What are BIM Dimensions, 2022; Habib, 2020).

Moreover, The growing construction trends and technologies have enabled the adoption of advanced approaches like BIM into the AEC industry that enables the benefits of realistic visualization, integration, and collaboration that ultimately gives sustainability outputs like efficient planning, conflict detection, waste management, design failure mitigation during construction, time, resource and cost efficiency, and improved productivity (Lee et al., 2021a). BIM can be optimized throughout the building's lifecycle. At the design phase of the building, it enables architectural and structural design analysis, mechanical, electrical, and plumbing design assessment, indoor environmental analysis, and energy consumption assessment. For the construction phase, it is beneficial for analyzing the construction progress as well as for post-construction evaluation. Moreover, it can be used for efficient control, management, and assessment of building operating and structural systems when integrated with IoT, ML, AI, and Digital twins (Panteli et al., 2020).

4.1.3. Building management system (BMS)

BMS is a centralized computer-based system that monitors and controls the electrical and mechanical systems or equipment of a building to validate sustainability (Hossain, 2019) (Al-Ghaili et al., 2021). These systems or equipment include Heating, Ventilation, Air conditioning (HVAC), lighting, plumbing, fire extinguishing, elevators, security, access control doors, closed-circuit television (CCTV), and contingency alarm systems (Minoli et al., 2017; Al-Ghaili et al., 2021;

Robotics in Renewable Energy, 2022). It can be remotely accessed and operated (Minoli et al., 2017). It helps in managing loads, optimizing desirable indoor conditions, decreasing energy consumption and carbon-related emissions, and minimizing building operational costs by operating building systems efficiently (Balaras, 2021). In essence, BMS can be integrated with other technologies like BIM or smart sensors to develop an intelligent building management system. Further, when coupled with the user interface, it potentially reduces energy consumption and enhances user comfort thereby improving overall efficiency through a cost-effective strategy that uses real-time data and simulations to optimize overall building performance (Habibi, 2017).

4.1.4. Smart meters

A smart meter is an electronic device that monitors the energy flow from or to the grid (electricity consumed from the grid or fed into the grid) and uses a form of electronic communication to transmit and receive data for information, monitoring, and control (Balaras, 2021). It enables two-way communication between supplier and consumer while measuring and recording the consumption of energy, water, and gas. It provides the user with real-time information on their domestic energy consumption like how much energy is being consumed and at what cost that enables them to take certain actions to minimize energy waste and consumption. Moreover, it can be monitored remotely (Kochañ et al., 2020). It helps in achieving financial incentives with timely monitoring and managing of energy consumption as stated in studies, smart metering implementation may account for up to 30% savings on energy bills for commercial buildings thereby improving the building's energy efficiency and sustainability (Rizvi, 2023). It may also be automated to redirect energy supply in case of any power excess or shortage enabling the building to be smart and sustainable (Avancini et al., 2019).

4.1.5. Smart grid

Simply defined, the Smart grid is the digitalization of conventional power grids (Isaksson et al., 2018). It is referred to as an evolving energy network of communication, controls, computers, advanced digital technologies, and equipment, working together to make the grid more reliable, secure, efficient, and greener (Charig, 2020). It enables two-way communication between the supplier/utility and

consumer/customer from energy generation, transmission, and distribution to consumption, to exchange, control, and monitor electricity and information flow as well as alter any change in energy supply and demand accordingly (Charig, 2020) (Balaras, 2021). Moreover, it integrates the regular power system with smart meters and controlling technologies that can facilitate renewable energy resources and decrease the pressure on power grids (Baidya et al., 2021). It integrates the needs and capabilities of all power generators, grid operators, consumers, and market stakeholders to optimize each part of the smart grid system as efficiently and sustainably as possible thereby improving its resilience, flexibility, and stability (Gonzalez, 2022). It enables the consumer to measure and record their real-time electricity consumption thereby saving energy and bills. Smart grid encompasses the potential to considerably reduce energy consumption, waste, greenhouse gas emissions, and carbon-related emissions (Diahovchenko et al., 2020).

4.1.6. Robotics

Robotics is a field of study that can be defined as an intersection of science, engineering, and technology that designs, constructs, and uses machines called robots to imitate or replicate the actions or activities of an intelligent creature, a human. Robots are programmable machines or devices that perform a specific task independently or under the controller's instructions (Daley, 2022) (European Construction Sector Observatory, 2021). More precisely, it can be referred to as an autonomous machine that possesses the capability to analyze its surrounding environment, carry out computations to make certain decisions and perform a specific task in the actual world (Guizzo, 2020).

Robots have been categorized into various types depending upon their features, capabilities, function, and area of application; Aerospace, Consumer, Disaster response, Drones/Aerial vehicles, Educational, Entertainment, Exoskeletons, Humanoids, Industrial, Medical, Military, Research, Self-driving cars, Augmenting, Software, Telepresence, and Construction robots. (Daley, 2022) (Guizzo, 2020) (Moravec, 2022). The development and implementation of robots for building construction can prove to be a disruptive trend for the building industry as it may enhance the productivity and efficiency of various construction activities (Afsari et al., 2018). Various robotic technologies have been developed including single-task robots (Drones) to integrated robots for on-site construction to perform construction activities more accurately, efficiently, and sustainably (Pan et al., 2020). Construction robots can deliver high-quality construction with precise and uniform work output, conduct difficult tasks in hazardous environments while eliminating safety risks for laborers, reduce construction time and resource consumption by reducing the possibility of mistakes, and eliminate repetitive tasks. Moreover, robots can be integrated with other technologies like Sensors, IOT, BIM, and drones to improve their performance efficiency (Digitalisation in the construction sector, 2021).

4.1.7. 3D printing/additive manufacturing/digital fabrication technology/rapid prototyping

3D printing technology which is also referred to as additive manufacturing or digital fabrication technology is an automated method to create/fabricate a physical object from a computer-aided geometrical representation like a CAD or BIM file (Durfee et al., 2019; Shahrubudin et al., 2019) (Digitalisation in the construction sector, 2021). It is done through layer-by-layer addition of materials (e.g. plastic, metals, concrete, resins, ceramics) (Ibrahim et al., 2022). The International Standards Organization (ISO) categorizes 3D printing into seven general types: Binder jetting, Material extrusion, Direct energy deposition, Powder bed fusion, Material jetting, Sheet lamination, and VAT polymerization. It's been opted and evolving for mass fabrication and production of open-source designs in different industries including Aerospace, Automotive, Robotics, Food, Agricultural, Healthcare, and AEC (Architecture, Building, and construction) industry (Shahrubudin et al., 2019).

For the building and construction industry 3D printing is primarily

used in the construction stage. Besides regular construction, it has been also used for the construction of energy-efficient and lightweight architectural and structural elements and the elements of unique shapes and complex geometries that are difficult to construct with traditional techniques. It possesses the potential to increase the project's productivity, reduce construction time, eliminate onsite accidents, reduce construction costs by opting for time and material-efficient machines, reduce construction and demolition waste, and minimize environmental pollution (Digitalisation in the construction sector, 2021). Several projects are underway to create cost-effective and time-efficient buildings with the 3D printer. One such project is Mvule Gardens in Kenya (Houser, 2023) and the standout aspect of this project is the cost as it has developed a 2-bedroom house with the approximate price of \$28,000 which is significantly lower than the average price of such a house e.g., approximately \$38,000. Moreover, the first 10 houses were manufactured/printed in merely 10 weeks.

4.1.8. Big data analytics

The term Big data refers to high volume, high variety, and high-velocity information assets or data that is collected from various sources like machines and sensors (The EU data protection reform, 2018). Big data technologies like BD analytics analyze these huge volumes of gathered data (Isaksson et al., 2018). Big data analytics is defined as a process, tool, application, or method that aims at structured or non-structured data collection, processing, cleaning and thereby analyzing to extract useful insights to make informed decisions from high-volume and high-velocity data sets that are collected from various sources (Colmenares-Quintero et al., 2021) (Plevris et al., 2022). Data mining, deep learning, and predictive analytics are some major methods of big data analysis. Applications of Big data analytics can be seen in any industry or area that deals with large volumes of data to get valuable information. Some of the applications are product development, supply-chain management, healthcare, sales and transactions, customer acquisition, and retention. Its incorporation can benefit cost reduction, product development, risk management, faster and better decision-making, and operational efficiency (Plevris et al., 2022). Big data technology applications can also be seen in the building and construction industry to achieve improved productivity and efficiency (Ram et al., 2019).

4.1.9. Internet of things

In literature, the concept of IoT is widely discussed from various perspectives. Krista Sorri (Sorri et al., 2022) identified and analyzed 122 various descriptions/definitions of IOT and further categorized them into 10 descriptive categories; Interaction, Virtual thing, Services, Physical object, Standardized technologies, Information, Data, Ubiquitous, User, and Uniqueness. R Minerva (Minerva et al., 2015) defined the IoT concept while considering earlier categories; "Internet of Things envisions a self-configuring, adaptive, complex network that interconnects 'things' to the Internet through the use of standard communication protocols. The interconnected things have physical or virtual representation in the digital world, sensing/actuation capability, and programmability features, and are uniquely identifiable. The representation contains information including the thing's identity, status, location, or any other business, social, or privately relevant information. These things offer services, with or without human intervention, through the exploitation of unique identification, data capture and communication, and actuation capability. The service is exploited through intelligent interfaces and is made available anywhere, anytime, and for anything considering security. Briefly defined IoT is defined as an internet-connected network of things or physical objects that relate to sensors and software that enable them to connect, transfer, and exchange data via wired or wireless networks with other systems and technologies to remotely communicate with each other (Lawal and Rafsanjani, 2021). There is a wide range of application areas of IoT e.g., smart cities, smart homes, and buildings, self-driving cars, smart grids

and energy management, healthcare, agriculture, retail and manufacturing, transportation, water and waste management, smart pollution control, and industrial automation are some areas (Terra, 2023).

For the building industry, IoT is implemented potentially through each phase, during the construction, it can be used to monitor workers' safety through a connected system of sensors. It enables remote monitoring of inaccessible and dangerous construction areas. Moreover, it has been reported that IoT integration in construction projects results in 22%–29% cost savings. During the operation and management of the building, IoT can be connected to building systems and devices for optimizing building control and monitoring, which may improve user comfort and save energy consumption. It has been reported that IoT may reduce 35% of energy consumption through coordinated digital building systems. Moreover, IoT can also help in construction and demolition waste management (Digitalisation in the construction sector, 2021).

4.1.10. Artificial intelligence

Artificial Intelligence is a disruptive technological discipline that develops intelligent machines and computers (Sharma et al., 2021). It can be defined as the science and engineering of developing computers and machines that can adapt/learn, reason and provide solutions through artificial neural networks and algorithms that would mimic human intelligence (What is Artificial Intelligence, 2022). Broadly defined, AI encompasses all kinds of methods and techniques that enable machines/computers to work and mimic human behavior to reproduce, execute, or even excel in human intelligence-related tasks with minimum or no intervention from humans (Janiesch et al., 2021) (Lyu et al., 2021).

AI covers numerous areas for its applications; cyber security, speech, and image recognition (automatic speech recognition (ASR), computer speech recognition), customer service (Virtual agents), predictive modeling, data analytics, recommendation engines, E-commerce, and financing, healthcare, agriculture, energy sector, smart homes, smart cities, Architecture, and Building industry (What is Artificial Intelligence, 2022) (Lee et al., 2021b) (Artificial intelligence, 2022) (Applications of Artificial Intelligence Across, 2023).

AI applications cover the life cycle of a building from planning and design, construction, operation, and maintenance to retrofit. Moreover, it may also ease the production of architectural design and visualization, Material selection and optimization, Structural design and analysis, Off-site manufacturing and automation, Construction management and progress monitoring, Risk management, Smart building operation, Building indoor environment control, Energy management and life cycle analysis (Baduge et al., 2022) (As et al., 2018) (Bingol et al., 2020) (Mehmood et al., 2019) (Seyedzadeh et al., 2018) (Ngarambe et al., 2020).

4.1.11. Machine learning

Machine learning can be described as a subfield of Artificial intelligence that equips the system with the ability to learn from extensive data to build an analytical model from that data or its representation to solve problems, give recommendations, make predictions, or give similar outcomes (Baduge et al., 2022). It enables the system to automatically learn on its own and improve its efficiency from experience while making the system from smart to smartest with time (Hong et al., 2020). It can further be categorized into supervised, semi-supervised, unsupervised, and reinforcement learning (Sarker, 2021).

Big Data, Advanced ML algorithms, and high-performance computing have facilitated the adoption and applications of machine learning in various areas (Yoon et al., 2020). Major application areas of ML include; Predictive analytics and intelligent decision-making, Cybersecurity and threat intelligence, Internet of things (IoT) and smart cities, Traffic prediction and transportation, Health sector and most recently for COVID-19 pandemic, E-commerce/retailer, Recommendation systems, Natural language processing (NLP) and sentiment

analysis, Face, image, speech and pattern recognition, Sustainable agriculture, User behavior analytics, Context-aware smartphone applications, building industry from planning and design to retrofitting and much more businesses and industries are transforming to ML technology (Yoon et al., 2020) (Sarker, 2021) (Sharma et al., 2021) (Chaudhary et al., 2020).

Buildings are considered to be one of the prime application areas for ML due to their ability to manage and enhance building performance throughout the building lifecycle. For the design stage, ML can be optimized for the generation and evaluation of the building's Architectural and structural design due to the accessibility of building codes and data (Yoon et al., 2020) (Tamke et al., 2018) (Má and laga-Chuquitaype, 2022). For building construction, it can be opted for Construction Management, construction cost analysis, retail price prediction, BIM, Digital Twin, waste management on site, construction material, and defect detection (Yoon et al., 2020) (Kim et al., 2013) (Rafiei et al., 2016) (Zabin et al., 2022) (Sacks et al., 2020). For operation, maintenance, and control, it can be optimized for fault detection and diagnostics (FDD), Energy efficiency enhancement to save energy, and improve comfort, and Post-occupancy evaluation which commonly covers the control and evaluation of Indoor environmental quality and Occupant's health, comfort, safety, and complaints (Yoon et al., 2020) (Pin et al., 2018). Even ML can be used at the retrofit stage to identify its potential and to evaluate energy conservation measures (Yoon et al., 2020).

4.1.12. Digital twin

A digital twin is a dynamic virtual replica of the real world's physical object, user, or activity in a digital world which is continuously been updated through data and information collected in real-time through smart sensors while keeping the historical data to make any prediction or decision through simulations, ML and reasoning (Coupry et al., 2021) (association, 2022). Moreover, the physical twin also imitates any change in the digital twin (Bortolini et al., 2022). More specifically for the building and construction industry, It combines design, construction, and operating data to replicate its physical twin e.g., actual building (Mangon, 2021). The most important part of the digital twin concept is that it needs to have a physical twin in the real world as without it (a physical twin), the Digital twin is nothing but a model (The untapped potential of virtual, 2021).

Digital twin enables designers to perform more simulations (Mangon, 2021). It helps to improve collaboration and control over the building operations as it simulates through collected data to make predictions and decisions based on actual world conditions. It eases the architects, designers, and engineers to meet the owner's requirements. Further, Fig. 7 depicts the concept of the digital twin while highlighting its significance for energy, operational, and cost efficiency with improved user comfort through real-time data exchange, monitoring, optimization, predictions, and informed decisions thereby leading toward sustainability (Energy and Carbon, 2022).

4.1.13. Blockchain

Blockchain is referred to as a decentralized database system, which serves as a digital ledger/log to store and chain together all transactions and activity records in the form of digital assets. Once the information is recorded, it becomes exceedingly difficult or nearly impossible to modify, cheat, or hack it. Digital assets are distributed and decentralized allowing for real-time transparency, control, and accessibility. The key features of this technology are digital ledger and immutability (Seebacher et al., 2017) (S, 2023).

Blockchain is an emerging technology for the built environment and its application can be seen in smart cities, smart homes, intelligent transportation, BIM, Digital twin, and construction management (Plevris et al., 2022). It has the potential to enable value in the built environment through smart contracts between owners and operators. It may enable the buildings of the future for machines to be designed and

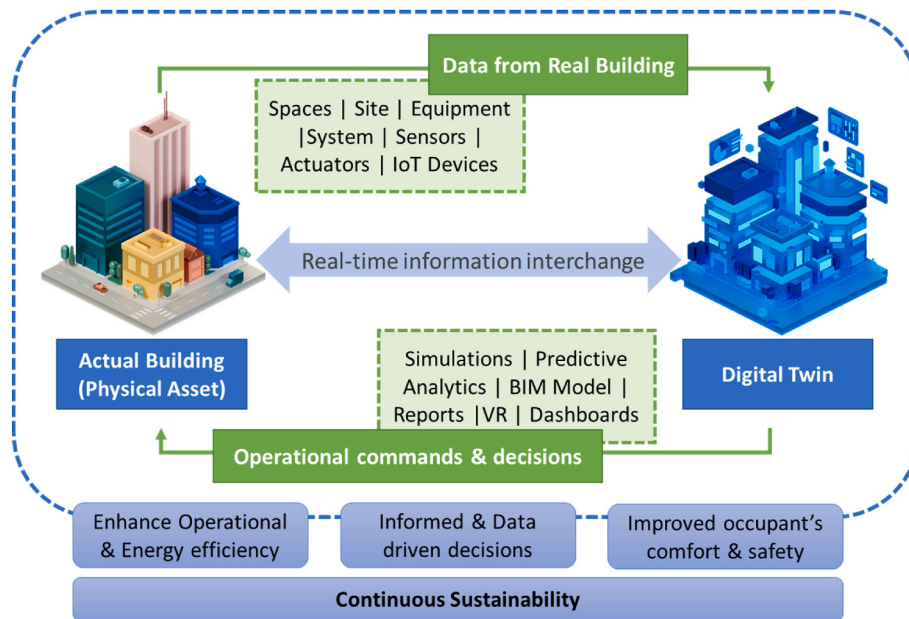


Fig. 7. Scope of Digital Twin for buildings.

integrated with buildings like sustainable micro-grid that would allow renewable energy to be generated, used, and stored on campus (Greaves). The project would manage its data more efficiently and sustainably, thus enabling uninterrupted collaboration among all stakeholders if Blockchain technology were employed from the project's inception. Moreover, buildings optimizing blockchain technology would be safer and data-rich with efficient risk management (Graham, 2019).

Further, Blockchain can be regarded as a key driver of sustainability in the building industry as it holds the potential to encourage the development of green, sustainable, and cost-effective buildings by enabling the efficient optimization and sustainable sourcing of construction materials, encouraging circular economy practices, promoting transparency and traceability throughout the supply chain and by providing real-time data for progress tracking and identifying issues (Team, 2022; Frą and kiewicz, 2023). More specifically, it has the potential to play a key role in the decarbonization of the building industry, which accounts for a large amount of the world's GHG emissions. It can be integrated with BIM throughout the lifecycle of a building to promote sustainability. It can collect environmental data and assist in the selection of low-carbon building materials during the planning stage. It can help with modeling energy-efficient building architecture in the design phase. It can enhance real-time stakeholder communication and logistics management during the construction phase. Lastly, blockchain-enabled energy management systems can monitor energy consumption during the operational phase, which makes it an effective tool for lowering emissions and enhancing efficiency (Reddy, 2023; Singh et al., 2023; Yang et al., 2023).

4.1.14. Cyber security

Cyber security refers to a set of technologies, processes, or practices to protect data, systems, networks, or sensitive information from unauthorized digital access or cyber-attacks (Vatuiu et al., 2019). Cyber security concept is becoming increasingly important as threats to digital assets are increasing with the continuous advancement and adoption of digital technologies. While digitalization offers advantages to industries, systems, and processes, it can also make them vulnerable to cyber-attacks from micro to macro levels. (IEA, 2017). For example, on one hand, the adoption of digital technologies in buildings like sensors, IoT, AI, and smart meters can substantially enhance the efficiency and sustainability of buildings, and on the other hand, the increased level of data collection, optimization, and connectivity may put them buildings

at risk of cyber-attacks (SIEMENS). While digitalization is transforming the building sector from smart homes to smart communities and smart cities – evident from examples like smart meters, micro-grids, and smart grids – the cyber security threats pose challenges at all levels, micro-, meso- — and macro. The more the building industry is digitalized, the more susceptible they are to cyber-attacks. Therefore, it is required to integrate cyber security in building projects, even at the planning and design stage of new buildings. Moreover, to ensure the cyber security of sustainable buildings, building managers and occupants must have awareness and training in building security while ensuring comprehensive data access and usage policies that would serve as a base for establishing cyber security plans. “Have a plan, make sure you understand the risks, have the proper controls, and ensure that they work. Doing this the right way is not hard.” Stated Matthew Clifford, head of energy and sustainability services—Asia Pacific at real estate and investment management firm JLL, while focusing on the importance of establishing cyber security systems for sustainable buildings (Kang and Controls, 2019).

5. Prospects of digital technologies

5.1. Adoption and market growth

The building industry is growing rapidly around the world and the adoption of digital technologies is continuously evolving with it. Building industry insight suggests that the digitalization-driven global smart building market size is expected to grow from around US\$ 97 billion in 2023 to over US\$ 408 billion by 2030. Such a significant growth is driven by the tangible benefits digital technologies are offering to buildings. An IBM report indicates that the implementation of digital technologies such as AI and IoT helped reduce buildings' energy use by 40% and maintenance costs by up to 30%. The digital technologies driving the smart building market growth are being broadly categorized under three types: building infrastructure management, security and emergency management, and energy management (Smart Building Market, 2023).

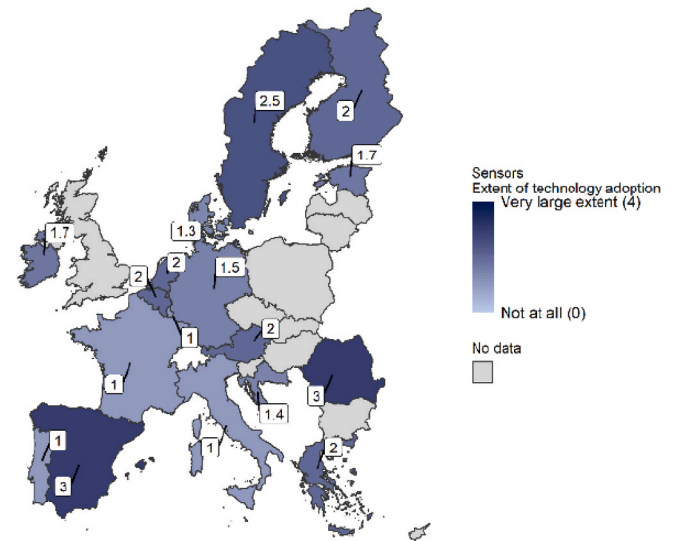
There have been significant technology-specific breakthroughs as well. For instance, Luyten, an Australian construction company, has developed the first 3D-printed home in the southern hemisphere that complies with building codes that took two days for printing, and one day for the assembly of printed elements. (FIRST 3D PRINTED HOUSE

IN, 2021). Table 2 highlights the adoption of digital technologies in Europe through construction companies which emphasizes BIM for highest adoption. The Middle Eastern construction sector is also experiencing a transformation with the optimization of disruptive digital technologies. As the PwC Middle East CP&I survey states technology adoption in the construction industry has crossed the 50% threshold, indicating a move toward more sophisticated solutions. Big Data, Drones, and Digital Twins are the three main areas of growth with adoption rates of 41%, 33%, and 17% respectively (PWC Together we build to, 2022; Disruptive technologies are transforming the, 2023).

The European market analysis highlights Sensors with the highest level of market maturity and adoption in the building industry due to their data acquisition capability. Especially for the construction stage, its adoption rate is growing rapidly (Digitalisation in the construction sector, 2021). The extent of Sensor adoption throughout Europe can be analyzed in Fig. 8. The adoption and research for Virtual sensors are still limited despite their enormous potential for smart and sustainable building applications (Yoon, 2022). The adoption of IoT with sensors is also advancing rapidly and it has been analyzed that building-related IoT has grown 21% from 2019 to 2021 and 12% annual growth is expected through 2027 (Marino, 2022).

BIM also exhibits a high adoption rate due to its efficient modeling and integration approach, but it is limited to only the design and construction phases of the project. Even though it is the most developed technology among others its usage is still limited in Europe as only 29% of building and construction companies work with BIM. Its global market is expected to grow by 18% from 2019 to 2027 and reach 23.2 billion USD with a compound annual growth rate (CAGR) of 17.2% (Digitalisation in the construction sector, 2021) (Soori et al., 2023). Digital twin is being rapidly used in different industries but it is evolving for the building sector. Its global market was at 3.1 billion USD in 2020 but is expected to project to 48.2 billion USD by 2026 and CAGR will rise 44% from 2022 to 2031 (association, 2022). AI and ML techniques are becoming popular due to their ability to analyze problems and make predictions, but their adoption is still less majorly limited to pilot studies and research, Fig. 9 highlights the AI adoption in the EU (Digitalisation in the construction sector, 2021). AI's global market for construction applications is expected to reach a CAGR of 33.8% from 2022 to 2027 (Zhang et al., 2022a).

The development of smart grids and smart grid technologies are being rapidly adopted and promoted around the globe due to their



Source: ECSO survey, 2020

Fig. 8. Level of sensor adoption in the EU (Christina Aprilia, 2022).

massive contribution towards sustainable development. Back in 2020, EU member states approved a 1.14 billion USD energy infrastructure project, from which 84% was reserved for smart grids and China Electricity Council reported a 0.63 trillion USD investment for smart grids (Gao et al., 2022). Smart meters being a prime component of the smart grid are also being deployed and various policies were made by different countries to boost their adoption in the building sector India launched Revamped Distribution Sector Scheme (RDSS) to install 25 crore smart meters by 2025, and European Commission reports that 225 million electrical smart meters and 51 million gas smart meters will be installed in the EU by 2024 leading to an estimation that 77% of European consumers will have electrical and 44% will have gas smart meters by 2024. The global market size of smart meters was reported to be USD 21.1 Billion in 2022 and is expected to rise with a CAGR of 9.4% from 2022 to 2032 (Kaitwade, 2022). The adoption of 3D printing has been rapidly growing in recent years - various reports and sources have indicated that the 3D printing global market for construction was around USD 18

Table 2

The digital technologies adoption by major European construction companies (Seebacher et al., 2017).

Country	Company	Revenue (CB)	BIM	Drones	VR	AR	3D Printing	AI & Big Data	IoT & Sensors	Robotics	Generative design
Spain	Acciona	7.3									
	ACS	34.9									
	FCC	5.8									
	Ferrovial	12.2									
UK	Balfour	7.9									
Netherland	BAM	6.6									
Belgium	BESIX	2.4									
France	Bouygues	32.9									
	Eiffage	15.3									
	Vinci	40.9									
Sweden	NCC	5.3									
	Skanska	15.3									
Austria	Strabag	13.5									

Daily business Multiple projects Single project/Prototype Research started No investment yet

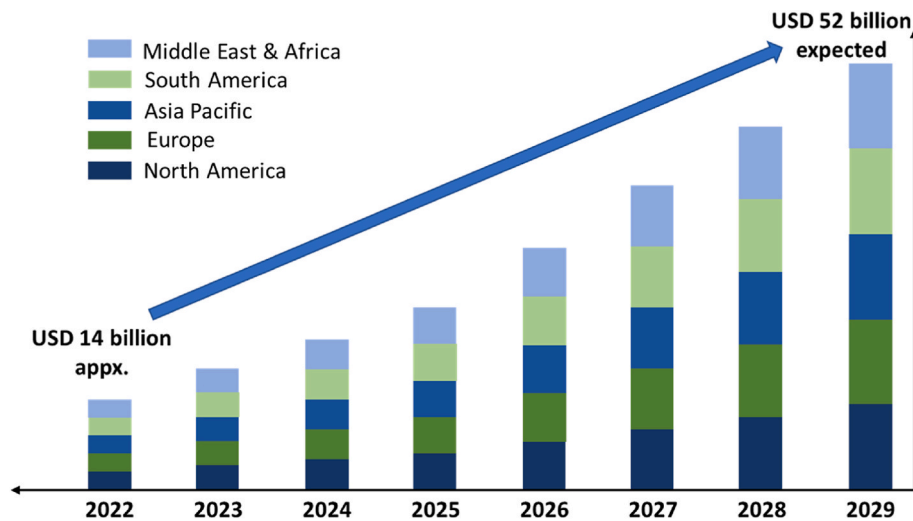


Fig. 11. Global BMS market (2019–2020) (Rumbens, 2023; MORGAN, 2023).

technologies bring about significant changes that frequently challenge established conventions and completely transform the way that buildings are planned, built, run, and maintained. Embracing the disruptive nature of digital technologies requires initiative, awareness and training, and inter-stakeholder engagement and cooperation besides regulatory frameworks. Similarly, Setaki, F. and A. van Timmeren (Setaki et al., 2022) also discussed the disruptive impact of digital technologies within the intersection of the circular economy and building industry, aiming to mitigate the sector's significant environmental footprint. Further, this study discussed the potential of digital technologies in revolutionizing the construction processes, waste management, and resource utilization within the building industry. By fundamentally transforming traditional practices, these technologies enable the adoption of circular principles, thereby facilitating the achievement of sustainability goals in the construction sector.

5.2.3. Compatibility

The application of digital technologies, especially in retrofitting, faces a major challenge in the form of compatibility. The functional characteristics of buildings' electrical and electronic architecture and infrastructure, and digital technologies may differ hugely which creates compatibility issues for digital technologies. While the compatibility issue is more common between different manufacturers the issue can also arise within the same manufacturer's products. The pace of technological and digital revolution is so rapid that often digital and control device systems from the same manufacturer are not compatible with only a few years old systems/appliances made by the same manufacturer. An important universal experience in this respect is the incompatibility of modern online meeting tools on devices that are only a few years old. This critical issue of compatibility requires an inter-stakeholder understanding especially involving regulators and manufacturers to find a way forward.

5.2.4. Cyber security

One of the biggest threats the digital revolution faces is the challenge of cyber security. Buildings, with the incorporation of different digital technologies especially with the likes of smart meters, BIM, BMS, and IoT, are also becoming prone to cyber threats. In the wake of trends like smart communities and smart cities, the issue of cyber security is critical at levels i.e., micro, meso- and macro-level building infrastructures. Some industry reports suggest that four out of 10 connected buildings can be at risk of cyber-attacks. The high risk can hamper the market growth of digital technologies (Smart Building Market, 2023). Buildings thus require to be equipped with strong cybersecurity measures and

contingent operational plans to protect sensitive data and critical systems. Data privacy is an important concern associated with digitalization. With the growing use of digital technologies, large data sets are being generated containing vital customer information including name, contact address, browsing history, and banking details besides other details like occupancy profile. The collection and use of data lead to concerns about data privacy and the safety and security of occupants and buildings. It is critical to ensure that data is collected and compiled in compliance with applicable laws and regulations (Disruptive technologies are transforming the, 2023).

Robust cyber security is, thus, vital to ensure the protection of private and sensitive data, the safety of consumers, and the safeguarding of critical infrastructure. The subject of cyber security has become across all sectors and the building sector can learn lessons from the experiences around. To begin with, there should be a thorough evaluation of cybersecurity risks to identify potential weak points, threats, and vulnerabilities in building systems, also recognizing the specific risks related to the employed digital systems in the building. Depending on the level of the incorporated digitalization, policies, and guidelines also need to be in place. There should be explicit cybersecurity rules and procedures for building administration encompassing guidelines for data protection, incident response, secure access control, and staff training. It should be backed with appropriate risk mitigation measures such as consumer and employee awareness and training, access control and authentication, network segmentation, regular software updates, employee training and awareness, encryption, incident response mechanism, regular security audits and testing, data backups and recovery, regulatory compliance, and continuous monitoring. As also reported by the U.S. Department of Energy (Reeve et al., 2020), there is an urgent need for robust cybersecurity measures in integrated building control systems. The increasing connectivity of these systems exposes them to significant cyber threats. Despite the potential benefits of interconnected technologies for energy efficiency and occupant comfort, vulnerabilities such as malware, phishing scams, and system failures pose substantial risks.

5.2.5. Skilled workforce

The lack of digital skills in the workforce is another important barrier to the adoption of digital technologies. Digital technologies require a new set of knowledge and skills to have them optimally operated and maintained. Improving worker digital skills is also essential from the perspective of cybersecurity. An educated and trained workforce is better able to recognize and react to potential threats and risks. Qualified and trained human resources are therefore part of the digital

transformation. The building sector needs to invest in training and professional development of their staff to enable them to effectively deal with digital technologies. Some of the action points in this respect may include developing cybersecurity awareness campaigns, establishing a proper cybersecurity training program, providing regular and updated training, simulating phishing exercises, certifications, and recognition, promoting an overall security-conscious culture, collaborating with external experts, and maintaining a system of feedback and continuous improvement. While digital technologies can enhance operational efficiency and reduce long-term costs, there are ongoing expenses for maintenance, updates, and training (Deloitte, 2023; Rumbens, 2023; MORGAN, 2023). Similarly, Safapour, E. and A. Silwal (Safapour and Silwal, 2023) also explored and reported the workforce-related challenges of digital technologies in the Architectural, Engineering, and Construction industries. It highlighted the lack of digital skills in the workforce as a significant barrier to the adoption of digital technologies. In addition, this study highlighted other workforce-related challenges such as routine-biased technological change, shifts in job market demand, an increase in the wage gap, and competency pressure.

5.2.6. Users' comprehension

The success of digital technologies in buildings also depends on user adoption. It is noteworthy that user-friendliness is important for any technology to create a positive user experience, drive adoption, reduce support costs, and gain a competitive edge. User's inability to understand digital technologies' functions and operational strategies undermines their effective utilization. Ease of user interface is therefore important.

Digitalization has become one of the most profound locally and globally integrating phenomena as it caters to the adoption and integration of digital technologies into various aspects of society, including the economy, governance, education, healthcare, and communication. It transcends geographical boundaries and connects individuals, institutions, businesses, and governments worldwide. Digitalization has facilitated the flow of data across borders, enabling global interconnectedness. Data is now seen as a valuable resource that drives innovation, research, and decision-making. Cross-border data flows support international collaborations, enable data-driven insights, and facilitate global integration of industries and sectors. While digitalization brings numerous benefits and opportunities for global integration, it also presents challenges, such as data privacy, cybersecurity, and the digital divide. Addressing these challenges requires international cooperation, collaboration, and the development of common frameworks and standards. The global nature of digitalization necessitates global integration efforts to harness its full potential and ensure its benefits are accessible to all.

6. Conclusions

Buildings, accounting for 40% of the world's energy consumption and 33% of greenhouse gas emissions, are required to play a leading role in the fight against climate change through extensive sustainability and smart initiatives. Following it, the building sector is adopting wide-ranging sustainability measures seeking greater energy and resource efficiency, and lower environmental emissions, and digitalization holds a significant potential in supporting these initiatives. Digitalization is revolutionizing the building industry by transforming the ways buildings are designed, constructed, operated, and managed. Research on digitalization is gaining momentum and following it, this study has reviewed and analyzed 13 prominent digital technologies that have significant potential to promote sustainability across the entire building lifecycle. The existing studies on the subject have limitations, mainly focusing on the application of specific technologies over a specific phase of the building lifecycle. A comprehensive account of a wide range of digital technologies covering the lifecycle of a building is missing in the literature which this study offers through a comprehensive analysis and

review of 13 major digital technologies considering the building lifecycle. This study is valuable also given the fact digitalization is rapidly evolving in terms of not only technological solutions but also applications in buildings. Given the broader digitalization trends, there is, thus, a greater and continuous need to more extensively study the prospects of digital technologies in the building sector. Against this backdrop, this study aims to bridge the gap in the literature by providing a comprehensive overview of digitalization in buildings in terms of technologies, their functions and applications, opportunities, challenges, and the way forward. Digital technologies are making buildings more integrated, flexible, energy-efficient, smart, and sustainable. The technologies covered in this study include sensors, building information modeling (BIM), Building Management Systems (BMS), smart meters, 3D printing/additive manufacturing, robotics, big data analytics, Internet of Things, machine learning, artificial intelligence, digital twin, blockchain, and cyber security.

Digital technologies offer wide-ranging functionality to buildings. Sensors, for example, play a pivotal role in the digitalization of buildings thereby enabling data collection, predictive modeling, and monitoring throughout a building's lifecycle while offering benefits in safety, efficiency, and maintenance. BIM can be employed throughout the building lifecycle holding significant potential for facilitating sustainability assessments and integration of advanced technologies, such as IoT, ML, AI, and Digital Twins, for efficient planning, design, construction, and management. BMS, centralized computer-based systems hold the potential of promoting sustainability by optimizing indoor conditions, reducing energy consumption, and lowering operational costs through integration with technologies like AI, BIM, and smart sensors. Smart meters and the smart grid contribute to enhanced energy efficiency and sustainability by providing real-time monitoring, automation, and optimization of energy consumption. Robotics are majorly used during the construction phase to perform tasks autonomously or under human control, enhancing construction efficiency, safety, and precision. 3D printing offers the benefits of potential cost savings, faster construction, reduced material waste, and hence sustainable practices in the building industry. Big data analytics offers insights to improve productivity, decision-making, and efficiency in the building and construction sector. IoT connects physical objects with sensors and software to exchange data, enabling applications such as safety monitoring during construction, energy optimization, and waste management in buildings. AI and ML hold the potential to revolutionize the building industry by optimizing design, construction, and operation processes, enhancing efficiency, and offering solutions from predictive modeling to energy management. Digital Twin enables real-time monitoring, predictive maintenance, and fault detection thereby offering simulations and enhanced control for improved building efficiency and sustainability. Blockchain enables transparency, traceability, and sustainable practices in the building industry, particularly in material sourcing, circular economy practices, and emissions reduction. Hence, these technologies are instrumental in promoting sustainability in buildings' lifecycle by optimizing resource utilization, enhancing operational efficiency, reducing energy consumption, minimizing waste, and fostering a holistic approach to building design, construction, operation, and management. With the generation, transaction, storage, and processing of large datasets, and networks, cyber security is becoming an integral part of digitalization to protect against unauthorized digital access or cyberattacks.

It is projected that over the 2018–2040 period, in the building sector, digital technologies can help improve the energy intensity of space heating, water heating, lighting, and cooking by 30%–50%. Digitalization technologies are experiencing rapid growth worldwide. The BMS market valued at USD 14 billion in 2021 is expected to reach USD 49 billion by 2030 at a CAGR of 14%. The BIM market is expected to grow by 18% between 2019 and 2027 to reach over USD 23 billion. Digital twin is projected to grow between 2020 and 2026 from USD 3.1 billion to USD 49 billion. 3D Printing's global market for construction was

around USD 18 billion in 2021/2022 and is expected to grow with CAGRs of 101.9% over the next decade. While digitalization offers numerous benefits for buildings, it also comes with several challenges that need to be addressed. These include high initial cost, privacy and security risks, and lack of customer awareness and skilled workforce. Digital technologies also face the issue of compatibility. The issues of cost and compatibility can be more critical in the case of retrofitting. These limiting factors, however, vary from technology to technology. It is crucial to conduct a thorough cost-benefit analysis to evaluate the specific needs, goals, and financial implications of incorporating digital technologies in a building project. It is important to enact policies that support digitalization. It should aim to foster innovation, promote investment, and create an enabling environment that supports the growth and adoption of digital technologies. To effectively address the digitalization challenges, legal and regulatory frameworks are required to be robustly established and continuously updated. This includes enacting laws that protect data privacy and security, regulate the collection and use of personal data, and establish guidelines for emerging technologies. Also, given the globalized scope of digitalization, there is a need to foster international cooperation to develop consistent and harmonized regulations to address global digital challenges. Data privacy and protection laws are still weak in some parts of the world. Governments and technology providers need to prioritize privacy and data protection by implementing strong data protection laws and regulations.

CRedit authorship contribution statement

Muhammad Asif: Writing – review & editing, Supervision, Methodology, Conceptualization. **Ghinwa Naem:** Writing – original draft, Data curation. **Muhammad Khalid:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgement

The work has been supported by the King Fahd University of Petroleum & Minerals, Saudi Arabia under the project No. DF201021.

References

- 3D Printing Construction Market Size, Share & Trends Analysis Report By Construction Method (Extrusion, Powder Bonding), By Material Type (Concrete, Metal), by End-User (Building, Infrastructure), and Segment Forecasts, 2023 – 2030, 2021.
- Afsari, K., et al., 2018. Applications of collaborative industrial robots in building construction. In: 54th ASC Annual International Conference Proceedings.
- Ahmed, W., Asif, M., 2021. A critical review of energy retrofitting trends in residential buildings with particular focus on the GCC countries. *Renew. Sustain. Energy Rev.* 144, 111000.
- Ahmed, W., Alazazmeh, A., Asif, M., 2023. Energy and water saving potential in commercial buildings: a retrofit case study. *Sustainability* 15 (1), 518.
- Al-Ali, A.-R., et al., 2017. A smart home energy management system using IoT and big data analytics approach. *IEEE Trans. Consum. Electron.* 63 (4), 426–434.
- Al-Ghaili, A.M., et al., 2021. Energy management systems and strategies in buildings sector: a scoping review. *IEEE Access* 9, 63790–63813.
- Alabbasi, M., Agkathidis, A., Chen, H., 2023. Robotic 3D printing of concrete building components for residential buildings in Saudi Arabia. *Autom. Construct.* 148, 104751.
- Alazazmeh, A., Asif, M., 2021. Commercial building retrofitting: assessment of improvements in energy performance and indoor air quality. *Case Stud. Therm. Eng.* 26, 100946.
- Alazazmeh, A., et al., 2022. Real-time data-based performance analysis of a large-scale building applied PV system. *Energy Rep.* 8, 15408–15420.
- Amaripadath, D., et al., 2023. Climate change sensitive sizing and design for nearly zero-energy office building systems in Brussels. *Energy Build.*, 112971.
- Anik, S.M.H., et al., 2022. A cost-effective, scalable, and portable IoT data infrastructure for indoor environment sensing. *J. Build. Eng.* 49, 104027.
- Applications of Artificial Intelligence across Various Industries, 2023. Available from: <https://www.forbes.com/sites/qai/2023/01/06/applications-of-artificial-intelligence/?sh=14892a663be4>.
- Artificial Intelligence (AI) for Cybersecurity, 2022. Available from: <https://www.ibm.com/security/artificial-intelligence>.
- As, I., Pal, S., Basu, P., 2018. Artificial intelligence in architecture: generating conceptual design via deep learning. *Int. J. Architect. Comput.* 16 (4), 306–327.
- Asif, M., 2022a. Handbook of Energy Transitions. CRC Press.
- Asif, M., 2022b. The 4Ds of Energy Transition: Decarbonization, Decentralization, Decreasing Use, and Digitalization. John Wiley & Sons.
- Asif, M., Ahmed, W., Alazazmeh, A., 2023. Energy performance assessment of a post-retrofit office building using measurement and verification protocol: a case study from KSA. *Energy Rep.* 9, 1366–1379.
- Association, W.N., 2022. Carbon dioxide emissions from electricity. Energy and Environment. <https://www.world-nuclear.org/information-library/energy-and-the-environment/carbon-dioxide-emissions-from-electricity.aspx>.
- Avancini, D.B., et al., 2019. Energy meters evolution in smart grids: a review. *J. Clean. Prod.* 217, 702–715.
- Baduge, S.K., et al., 2022. Artificial intelligence and smart vision for building and construction 4.0: machine and deep learning methods and applications. *Autom. Construct.* 141, 104440.
- Baidya, S., et al., 2021. Reviewing the opportunities, challenges, and future directions for the digitalization of energy. *Energy Res. Social Sci.* 81, 102243.
- Balaras, C., 2021. Energy and Digitalization in the Building Sector.
- Bingol, K., et al., 2020. Artificial Intelligence Applications in Earthquake Resistant Architectural Design: Determination of Irregular Structural Systems with Deep Learning and ImageAI Method.
- Bortolini, R., et al., 2022. Digital twins' applications for building energy efficiency: a review. *Energies* 15 (19), 7002.
- Bouckaert, S., et al., 2021. Net Zero by 2050: A Roadmap for the Global Energy Sector.
- Brenner, B., Hartl, B., 2021. The perceived relationship between digitalization and ecological, economic, and social sustainability. *J. Clean. Prod.* 315, 128128.
- Building Management System Market Size and Forecast, 2023.
- Building the Future: the Role of Digital Adoption and Integration for Construction Resilience, 2023. Available from: <https://www.planradar.com/au/building-the-future-digital-adoption-for-construction/>.
- Bumann, J., Peter, M., 2019. Action fields of digital transformation—a review and comparative analysis of digital transformation maturity models and frameworks. *Digitalisierung und andere Innovations. Manag.* 2, 13–40.
- Charig, N., 2020. The Smart Grid – what it Is, and Why We Need it.
- Chaudhary, S., et al., 2020. A brief review of machine learning and its applications. *J. Phys. Sci., Eng. Technol.* 12 (Suppl. 1), 218–223.
- Chen, Y., et al., 2022. Construction 4.0, industry 4.0, and building information modeling (BIM) for sustainable building development within the smart city. *Sustainability* 14 (16), 10028.
- Choi, Y., Yoon, S., 2023. In-situ observation virtual sensor in building systems toward virtual sensing-enabled digital twins. *Energy Build.* 281, 112766.
- Christina Aprilia, R.J.P.S., 2022. How Digitalized Buildings Can Improve Energy Efficiency in ASEAN. ASEAN Center for Energy.
- Colmenares-Quintero, R.F., et al., 2021. Big Data analytics in Smart Grids for renewable energy networks: systematic review of information and communication technology tools. *Cogent Eng.* 8 (1), 1935410.
- Coupry, C., et al., 2021. BIM-Based digital twin and XR devices to improve maintenance procedures in smart buildings: a literature review. *Appl. Sci.* 11 (15), 6810.
- Daley, S., 2022. Robotics Technology. Available from: <https://builtin.com/robotics>.
- Debrah, C., Chan, A.P., Darko, A., 2022. Artificial intelligence in green building. *Autom. Construct.* 137, 104192.
- Deloitte, 2023. The State of Digital Adoption in Construction Report 2023-Autodesk.
- Diahovchenko, I., et al., 2020. Progress and challenges in smart grids: distributed generation, smart metering, energy storage and smart loads. *Iran. J. Sci. Technol. Trans. Electric. Eng.* 44, 1319–1333.
- DigiFloat-Principle power. Available from: <https://www.principlepower.com/expanding-frontiers/digifloat>.
- Digital Transformation in the Building Sector, 2022. Available from: <https://bau-muenchen.com/en/supporting-program/key-themes/digital-transformation/>.
- Digitalisation in the construction sector, 2021. European Construction Sector Observatory.
- Digitalization - The Key Enabler of Smart, 2019. Buildings and Cities, 2 September 2019, Available from: <https://buildings.honeywell.com/gb/en/news-events/news/2019/09/digitalization-the-key-enabler-of-smart-buildings-and-cities>.
- Disruptive Technologies Are Transforming the Middle East Construction Sector-PWC, 2023. PWC.
- Dörfler, K., et al., 2022. Additive Manufacturing using mobile robots: opportunities and challenges for building construction. *Cement Concr. Res.* 158, 106772.
- Durfée, W.K., Iazzo, P.A., 2019. Medical applications of 3D printing. In: *Engineering in Medicine*. Elsevier, pp. 527–543.
- Energy, Carbon & Sustainability Digital Twin: Generic Functional Specification, 2022. Energy Efficiency 2019 (The Authoritative Tracker of Global Energy Efficiency Trends), 2019.
- European Construction Sector Observatory: Digitalisation in the Construction Sector, 2021.

- FIRST 3D PRINTED HOUSE IN AUSTRALIA First Code Compliant 3D Printed House In Australia and Southern Hemisphere, 2021. Available from: <https://www.luyten3d.com/first-3dprinted-house-in-australia-heptapod>.
- Follini, C., et al., 2020. Bim-integrated collaborative robotics for application in building construction and maintenance. *Robotics* 10 (1), 2.
- Four Ways Smart Building Technology Creates Sustainable Environments, 2021.
- Frąckiewicz, M., 2023. Blockchain for sustainable construction: how it helps to improve traceability and efficiency. Available from: <https://ts2.space/en/blockchain-for-sustainable-construction-how-it-helps-to-improve-traceability-and-efficiency/#:~:text=By%20using%20a%20blockchain%2Dbased,more%20efficient%20and%20cost%20effective>.
- Frenzel, A., et al., 2021. Digitization or digitalization? Toward an understanding of definitions, use and application in IS research. In: AMCIS.
- Gao, Y., Fang, C., Zhang, J., 2022. A spatial analysis of smart meter adoptions: empirical evidence from the US data. *Sustainability* 14 (3), 1126.
- García-Moreno, S., López-Ruiz, V.-R., 2023. A review of the energy sector as a key factor in industry 4.0: the case of Spain, 16 (11), 4446.
- Gonzalez, P., 2022. Smart Grids.
- Graham, S., 2019. Implementation of Blockchain Technology in the Construction Industry.
- IEA, Buildings, International Energy Agency, 2022, Buildings - Energy System - IEA.
- Greaves, P. Blockchain's Impact on Buildings.
- Guizzo, E., 2020. ROBOTS-YOUR GUIDE TO THE WORLD OF ROBOTICS. Available from: <https://robots.ieee.org/learn/what-is-a-robot/>.
- Habib, H.M., 2020. Employ 6D-BIM model features for buildings sustainability assessment. In: IOP Conference Series: Materials Science and Engineering. IOP Publishing.
- Habibi, S., 2017. Micro-climatization and real-time digitalization effects on energy efficiency based on user behavior. *Build. Environ.* 114, 410–428.
- Handbook of Energy and Environmental Security, 2022. Academic Press.
- Hong, T., et al., 2020. State-of-the-art on research and applications of machine learning in the building life cycle. *Energy Build.* 212, 109831.
- Hossain, M., 2019. Sustainable Design and Build: Building, Energy, Roads, Bridges, Water and Sewer Systems.
- Houser, K., 2023. World's largest 3D-printed affordable housing project launches in Kenya. Available from: <https://www.freethink.com/cities/3d-printed-affordable-housing>.
- Huovila, P., et al., 2013. Buildings and Climate Change: Summary for Decision-Makers.
- Ibrahim, I., et al., 2022. 3D printing in sustainable buildings: systematic review and applications in the United Arab Emirates. *Buildings* 12 (10), 1703.
- IEA, P., 2017. Digitalisation and Energy.
- Ili, S., 2019. AI in buildings: companies need to rethink their business models. Available from: <https://www.siemens.com/global/en/company/stories/infrastructure/2019/digitalization-and-ai-in-the-building-sector.html>.
- Iqbal, J., et al., 2019. Robotics inspired renewable energy developments: prospective opportunities and challenges. *IEEE Access* 7, 174898–174923.
- Isaksson, A.J., Harjunoski, I., Sand, G., 2018. The impact of digitalization on the future of control and operations. *Comput. Chem. Eng.* 114, 122–129.
- Janiesch, C., Zschech, P., Heinrich, K., 2021. Machine learning and deep learning. *Electron. Mark.* 31 (3), 685–695.
- Kaewunruen, S., Rungskunroch, P., Welsh, J., 2018. A digital-twin evaluation of net zero energy building for existing buildings. *Sustainability* 11 (1), 159.
- Kaitwade, N., 2022. Smart Meter Market.
- Kang, M., 2019. In: Controls, Johnson (Ed.), Smart and Sustainable, but Are These Green Buildings Secure?. *Eco Business*.
- Kaufmann, D., Ruaux, X., Jacob, M., 2018. Digitalization of the Construction Industry: the Revolution Is Underway. Oliver Wyman.
- Khajavi, S.H., et al., 2019. Digital twin: vision, benefits, boundaries, and creation for buildings. *IEEE Access* 7, 147406–147419.
- Kim, G.-H., et al., 2013. Comparison of school building construction costs estimation methods using regression analysis. *Neural Network, and Support Vector Machine*.
- Kochański, M., Korczak, K., Skoczowski, T., 2020. Technology innovation system analysis of electricity smart metering in the European Union. *Energies* 13 (4), 916.
- Kumar, T., Srinivasan, R., Mani, M., 2022. An emery-based approach to evaluate the effectiveness of integrating IoT-based sensing systems into smart buildings. *Sustain. Energy Technol. Assessments* 52, 102225.
- Kumar Singh, A., et al., 2023. Investigating the barriers to the adoption of blockchain technology in sustainable construction projects. *J. Clean. Prod.* 403, 136840.
- Kumara, W., Waidyasekara, K., Weerasinghe, R., 2016. Building management system for sustainable built environment in Sri Lanka. *Built. Environ. Proj. Asset. Manag.* 6 (3), 302–316.
- Lawal, K., Rafsanjani, H.N., 2021. Trends, benefits, risks, and challenges of IoT implementation in residential and commercial buildings. Vol. 3, Issue 3, *Energy Built Environ.*
- Lee, J.Y., Irisboev, I.O., Ryu, Y.-S., 2021a. Literature review on digitalization in facilities management and facilities management performance measurement: contribution of industry 4.0 in the global era. *Sustainability* 13 (23), 13432.
- Lee, D., Yoon, S.N., 2021b. Application of artificial intelligence-based technologies in the healthcare industry: opportunities and challenges. *Int. J. Environ. Res. Publ. Health* 18 (1), 271.
- Li, J., Greenwood, D., Kassem, M., 2018. Blockchain in the Built Environment: Analysing Current Applications and Developing an Emergent Framework. Diamond Congress Ltd., Budapest University of Technology and Economics.
- Lokshina, I.V., Greguš, M., Thomas, W.L., 2019. Application of integrated building information modeling, IoT and blockchain technologies in system design of a smart building. *Procedia Comput. Sci.* 160, 497–502.
- Lyu, W., Liu, J., 2021. Artificial Intelligence and emerging digital technologies in the energy sector. *Appl. Energy* 303, 117615.
- Málaga-Chuquityape, C., 2022. Machine learning in structural design: an opinionated review. *Front. Built Environ.* 6.
- Ma, L., Hartmann, T., 2023. A proposed ontology to support the hardware design of building inspection robot systems. *Adv. Eng. Inf.* 55, 101851.
- Magazine, B., 2022. Digitalization: the future of building technology. Available from: <https://www.bfmmagazine.co.uk/digitalization-the-future-of-building-technology/>.
- Malik, A., et al., 2022. 3D printing towards implementing Industry 4.0: sustainability aspects, barriers and challenges. *Ind. Robot. Int. J. Robot. Res. Appl.* 49 (3), 491–511.
- Mangon, N., 2021. What is digitalization? Here comes the Fourth transformation in building design. Available from: <https://redshift.autodesk.com/articles/what-is-digitalization>.
- Marino, A., 2022. How AI and ML Can Reduce Energy Demand to Help Buildings Improve Long-Term Sustainability and Cost Savings. Schneider Electric Blog.
- Market and Market, 2023. Smart Meter Market.
- Martin, D., Kühl, N., Satzger, G., 2021. Virtual sensors. *Bus. Inf. Syst. Eng.* 63, 315–323.
- Marzouk, M., Atef, M., 2022. Assessment of indoor air quality in academic buildings using IoT and deep learning. *Sustainability* 14 (12), 7015.
- Mehmood, M.U., et al., 2019. A review of the applications of artificial intelligence and big data to buildings for energy-efficiency and a comfortable indoor living environment. *Energy Build.* 202, 109383.
- Merimi, M., Taghipour, A., 2021. Accelerating the digitalization of the supply chain: an empirical research about COVID-19 crisis. In: *Digitalization of Decentralized Supply Chains during Global Crises*. IGI Global, pp. 1–24.
- Metwally, E.A., Farid, A.A., Ismail, M.R., 2022. Development of an IoT assessment method: an interdisciplinary framework for energy efficient buildings. *Energy Build.* 254, 111545.
- Minerva, R., Biru, A., Rotondi, D., 2015. Towards a definition of the internet of things (IoT). *IEEE Internet Initiat.* 1 (1), 1–86.
- Minoli, D., Sohraby, K., Occhiogrosso, B., 2017. IoT considerations, requirements, and architectures for smart buildings—energy optimization and next-generation building management systems. *IEEE Internet Things J.* 4 (1), 269–283.
- Moravec, H.P., 2022. Robot technology. Available from: <https://www.britannica.com/technology/robot-technology>.
- Morgan, S., 2023. Construction's big challenge: tech adoption. In: *Digital Construction Week*.
- Ngarambe, J., Yun, G.Y., Santamouris, M., 2020. The use of artificial intelligence (AI) methods in the prediction of thermal comfort in buildings: energy implications of AI-based thermal comfort controls. *Energy Build.* 211, 109807.
- Njuguna Ndung'u, L.S., 2020. The Fourth Industrial Revolution and Digitization Will Transform Africa into a Global Powerhouse.
- Olu-Ajayi, R., et al., 2022. Building energy consumption prediction for residential buildings using deep learning and other machine learning techniques. *J. Build. Eng.* 45, 103406.
- Pan, M., et al., 2020. Influencing factors of the future utilisation of construction robots for buildings: a Hong Kong perspective. *J. Build. Eng.* 30, 101220.
- Pandey, A., Asif, M., 2022. Assessment of energy and environmental sustainability in south Asia in the perspective of the sustainable development goals. *Renew. Sustain. Energy Rev.* 12492.
- Panteli, C., Kyllil, A., Fokaides, P.A., 2020. Building information modelling applications in smart buildings: from design to commissioning and beyond A critical review. *J. Clean. Prod.* 265, 121766.
- Perera, S., et al., 2023. Drivers and barriers to digitalisation: a cross-analysis of the views of designers and builders in the construction industry. *J. Inf. Technol. Construct.* 28 (5), 87–106.
- Pham, A.-D., et al., 2020. Predicting energy consumption in multiple buildings using machine learning for improving energy efficiency and sustainability. *J. Clean. Prod.* 260, 121082.
- Phillips, R., Fannon, D., Eckelman, M., 2022. Dynamic modeling of future climatic and technological trends on life cycle global warming impacts and occupant satisfaction in US office buildings. *Energy Build.* 256, 111705.
- Pin, C., Medina, C., McArthur, J., 2018. Supporting post-occupant evaluation through work order evaluation and visualization in FM-BIM. In: *ISARC. Proceedings of the International Symposium on Automation and Robotics in Construction*. IAARC Publications.
- Plevris, V., Lagaros, N.D., Zeytinci, A., 2022. Blockchain in civil engineering, architecture and construction industry: state of the art, evolution, challenges and opportunities. *Front. Built Environ.* 49.
- Pradeep, K., Venkatesan, S., Rajan, S., 2021. A neural network based smart building monitoring system using wireless sensor network. *Inf. Technol. Ind.* 9 (1), 232–239.
- PWC Together We Build to Transform Societies the 2022 Middle East Capital Projects And Infrastructure Survey, 2022.
- Quadrat-Ullah, H., 2020. Introduction: dynamics of energy, environment, and economy; A sustainability perspective. *Dyn. Energy, Environ. Econ.: Sustain. Perspect.* 3–14.
- Rafiei, M.H., Adeli, H., 2016. A novel machine learning model for estimation of sale prices of real estate units. *J. Construct. Eng. Manag.* 142 (2), 04015066.
- Ram, J., Afridi, N.K., Khan, K.A., 2019. Adoption of Big Data analytics in construction: development of a conceptual model. *Built. Environ. Proj. Asset. Manag.* 9 (4), 564–579.
- Razali, M.F., et al., 2019. A review: application of Building Information Modelling (BIM) over building life cycles. In: *IOP Conference Series: Earth and Environmental Science*. IOP Publishing.
- Reddy, K., 2023. Decarbonizing the Built Environment with Blockchain.

- Reeve, H., et al., 2020. Challenges And Opportunities To Secure Buildings From Cyber Threats. U.S.Department of Energy.
- Rizvi, M.R., 2023. Smart metering for commercial buildings. In: Achieving Energy Efficiency and Cost Savings with Specific Examples. Available from: <https://medium.com/@mrizvi0310/smart-metering-for-commercial-buildings-achieving-energy-efficiency-and-cost-savings-with-specific-ddb20f7a7c4>.
- Robotics in Renewable Energy, 2022. Robotics. Available from: <https://whatnextglobal.com/robotics-in-renewable-energy/>.
- Rumbens, D., 2023. The State of Digital Adoption in Construction Report 2023. Available from: <https://www.deloitte.com/au/en/services/economics/analysis/state-digital-adoption-construction-report-2023.html>.
- S, R.A., 2023. What is blockchain technology? How does blockchain work? Available from: <https://www.simplilearn.com/tutorials/blockchain-tutorial/blockchain-technology>.
- Sacks, R., et al., 2020. Construction with digital twin information systems. *Data-Centric Eng.* 1, e14.
- Safapour, E., Silwal, A., 2023. Workforce-Related Challenges of Construction 4.0 Technologies in the Architectural, Engineering, and Construction Industry. <https://doi.org/10.1061/9780784485231.074>.
- Sakin, M., Kiroglu, Y.C., 2017. 3D printing of buildings: construction of the sustainable houses of the future by BIM. *Energy Proc.* 134, 702–711.
- Sarker, I.H., 2021. Machine learning: algorithms, real-world applications and research directions. *SN Comput. Sci.* 2 (3), 160.
- Sategna, L.G., Meinerio, D., Volontà, M., 2019. Digitalising the Construction Sector, Unlocking the Potential of Data with a Value Chain Approach. CECE, Innovation and Development Consulting Sprl, S.I.R.I.O. Srl.
- Sayadi, S., et al., 2022. Analyzing the climate-driven energy demand and carbon emission for a prototype residential nZEB in central Sweden. *Energy Build.* 261, 111960.
- Seebacher, S., Schüritz, R., 2017. Blockchain technology as an enabler of service systems: a structured literature review. In: Exploring Services Science: 8th International Conference, IESS 2017, Rome, Italy, May 24–26, 2017, Proceedings 8. Springer.
- Setaki, F., van Timmeren, A., 2022. Disruptive technologies for a circular building industry. *Build. Environ.* 223, 109394.
- Seyedzadeh, S., et al., 2018. Machine learning for estimation of building energy consumption and performance: a review. *Vis. Eng.* 6, 1–20.
- Shahrubudin, N., Lee, T.C., Ramlan, R., 2019. An overview on 3D printing technology: technological, materials, and applications. *Procedia Manuf.* 35, 1286–1296.
- Sharma, N., Sharma, R., Jindal, N., 2021. Machine learning and deep learning applications-a vision. *Glob. Transit. Proc.* 2 (1), 24–28.
- Shojaei, A., Wang, J., Fenner, A., 2019. Exploring the feasibility of blockchain technology as an infrastructure for improving built asset sustainability. *Built. Environ. Proj. Asset. Manag.*
- SIEMENS. Cybersecurity for smart buildings. Available from: <https://www.siemens.com/global/en/products/buildings/cybersecurity-for-smart-buildings.html>.
- Singh, R., et al., 2021. Cloud manufacturing, internet of things-assisted manufacturing and 3D printing technology: reliable tools for sustainable construction. *Sustainability* 13 (13), 7327.
- Singh, A.K., et al., 2023. A strategic roadmap to overcome blockchain technology barriers for sustainable construction: a deep learning-based dual-stage SEM-ANN approach. *Technol. Forecast. Soc. Change* 194, 122716.
- FBI Smart Building Market, 2023.
- Soori, M., Arezoo, B., Dastres, R., 2023. Internet of things for smart factories in industry 4.0, a review. *Internet Things Cyber-Phys. Syst.* 3, 192–204.
- Sorri, K., Mustafee, N., Seppänen, M., 2022. Revisiting IoT definitions: a framework towards comprehensive use. *Technol. Forecast. Soc. Change* 179, 121623.
- Tamke, M., Nicholas, P., Zwierzycki, M., 2018. Machine learning for architectural design: practices and infrastructure. *Int. J. Architect. Comput.* 16 (2), 123–143.
- Team, F.M., 2022. Digitalisation and Blockchain as Sustainability Drivers in the Construction Industry.
- Terra, J., 2023. Real-World IoT Applications in 2023.
- The EU Data Protection Reform and Big Data (Fact sheet), 2018. <https://doi.org/10.2838/190200>.
- The Untapped Potential of Virtual Sensors for Digital Twins, 2021. Available from: <https://www.bluefruit.co.uk/quality/virtual-sensors-for-digital-twins/>.
- Thelen, D., Zijlstra, R., Zandbergen, Z., 2021. Digitalization of the Built Environment: towards a More Sustainable Construction Sector. World Business Council for Sustainable Development, Geneva.
- Tricoire, J.-P., 2021. Why buildings are the foundation of an energy-efficient future. Available from: <https://www.weforum.org/agenda/2021/02/why-the-buildings-of-the-future-are-key-to-an-efficient-energy-ecosystem/>.
- Vatui, T., Lazaroiu, G., 2019. Using renewable energy sources in the context of promoting a conceptual model for sustainable cloud computing. *Survey. Geol. Mining Ecol. Manag. (SGEM)*: Sofia 593–601.
- What Are BIM Dimensions – 3D, 4D, 5D, 6D, and 7D BIM?, 2022.
- Available from: What Is Artificial Intelligence (AI)?, 2022 <https://www.ibm.com/topics/artificial-intelligence>.
- Xiang, Y., et al., 2022. Research on sustainability evaluation of green building engineering based on artificial intelligence and energy consumption. *Energy Rep.* 8, 11378–11391.
- Yang, Z., et al., 2023. Blockchain technology in building environmental sustainability: a systematic literature review and future perspectives. *Build. Environ.* 245, 110970.
- Yevu, S.K., Yu, A.T.W., Darko, A., 2021. Digitalization of construction supply chain and procurement in the built environment: emerging technologies and opportunities for sustainable processes. *J. Clean. Prod.* 322, 129093.
- Yoon, S., 2022. Virtual sensing in intelligent buildings and digitalization. *Autom. Construct.* 143, 104578.
- Yoon, S.-H., et al., 2018. Multiple power-based building energy management system for efficient management of building energy. *Sustain. Cities* 42, 462–470.
- Yoon, S., et al., 2020. Virtual sensors for estimating district heating energy consumption under sensor absences in a residential building. *Energies* 13 (22), 6013.
- Zabin, A., et al., 2022. Applications of machine learning to BIM: a systematic literature review. *Adv. Eng. Inf.* 51, 101474.
- Zhang, F., et al., 2022a. Integrated applications of building information modeling and artificial intelligence techniques in the AEC/FM industry. *Autom. Construct.* 139, 104289.
- Zhang, F., et al., 2022b. Integrated Applications of Building Information Modeling and Artificial Intelligence Techniques in the AEC/FM Industry, vol. 139, 104289.
- Zhao, Y., Genovese, P.V., Li, Z., 2020. Intelligent thermal comfort controlling system for buildings based on IoT and AI. *Future Internet* 12 (2), 30.